



Critical Design Review Document



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1 Introduction

The INSPACe CANSAT Competition serves as a platform for aspiring engineers, scientists, and space enthusiasts to apply their technical knowledge in a practical, real-world aerospace challenge. This competition fosters innovation, collaboration, and problem-solving by tasking teams with designing, building, and launching a miniature satellite, or CANSAT, to meet a set of mission objectives.

This Complete Design Review (CDR) Document encapsulates our team's design, development, and testing processes. It serves as a comprehensive record of our mission objectives, system architecture, payload design, electronics integration, software development, and flight readiness. The document also details the challenges encountered, the solutions implemented, and the validation methods used to ensure a successful mission.

Our approach to the CANSAT design prioritizes functionality, reliability, and innovation while adhering to the competition guidelines. Each subsystem has been meticulously designed and tested to ensure optimal performance in the demanding conditions of launch and recovery. Additionally, we have emphasized teamwork and interdisciplinary collaboration, ensuring that our final product is a well-engineered solution that meets the expectations of the competition.

Through this journey, we have gained invaluable insights into aerospace engineering, mission planning, and systems engineering. This document not only reflects our technical expertise but also showcases our passion for space exploration and our commitment to learning. We look forward to contributing to the broader scientific community and leveraging this experience for future advancements in aerospace technology.

Regards,
Team Brahmand,
IIIT-Hyderabad

2 Team Description



 **Harikumar Kandath**
Faculty Mentor

 **Harsh Kapoor**
Team Lead

Mechanical Team

 **Chamarthy Madhan Sai Krishna**
Team Coordinator

 **Vignesh Vembar Manikantan**
Team Member

 **Vijay S.R. Aravyanthan**
Team Member

 **Harsh Kapoor**
Team Lead

Electrical & Sensor Team

 **Adithya Kishor**
Team Member

 **Aditya R. Shankar**
Team Member

 **Anushka Sharma**
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 **Adithya Kishor**
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FSW & GCS Team

 **Adithya Kishor**
Team Member

 **Vijay S.R. Aravyanthan**
Team Member

Management Team

 **Chamarthy Madhan Sai Krishna**
Team Coordinator

 **Harsh Kapoor**
Team Lead

3 List of Acronyms

Acronym	Full Form	Acronym	Full Form
CANSAT	Can Satellite	INSPACe	Indian National Space Promotion and Authorization Center
GPS	Global Positioning System	IMU	Inertial Measurement Unit
PCB	Printed Circuit Board	RF	Radio Frequency
ADC	Analog to Digital Converter	MCU	Microcontroller Unit
PWM	Pulse Width Modulation	ISP	In-System Programming
FSW	Flight Software	GCS	Ground Control Station
LEO	Low Earth Orbit	EMI	Electromagnetic Interference
RCS	Reaction Control System	SLA	Stereo Lithography Apparatus
CONOPS	Concept of Operations	VHF	Very High Frequency
Li-Po	Lithium Polymer (Battery)	ESC	Electronic Speed Controller
A	Ampere	m	meter
°C	Celsius	mA	Milli-Ampere
dB _i	decibel	mm	milli meter
g	Acceleration due to gravity	mW	milli Watt
G	G-force	MB	Mega Byte
GHz	Giga-Hertz	MHz	Mega-Hertz
h	hour	μA	Micro-Ampere
hPa	hecta Pascal	N	Newton
Hz	Hertz	Pa	Pascal
km	kilometer	s	second
kPa	Kilo Pascal	V	Volt
KB	Kilo Byte	Kg	Kilo gram

Table 1: Abbreviations and their Full Forms

4 Systems Overview

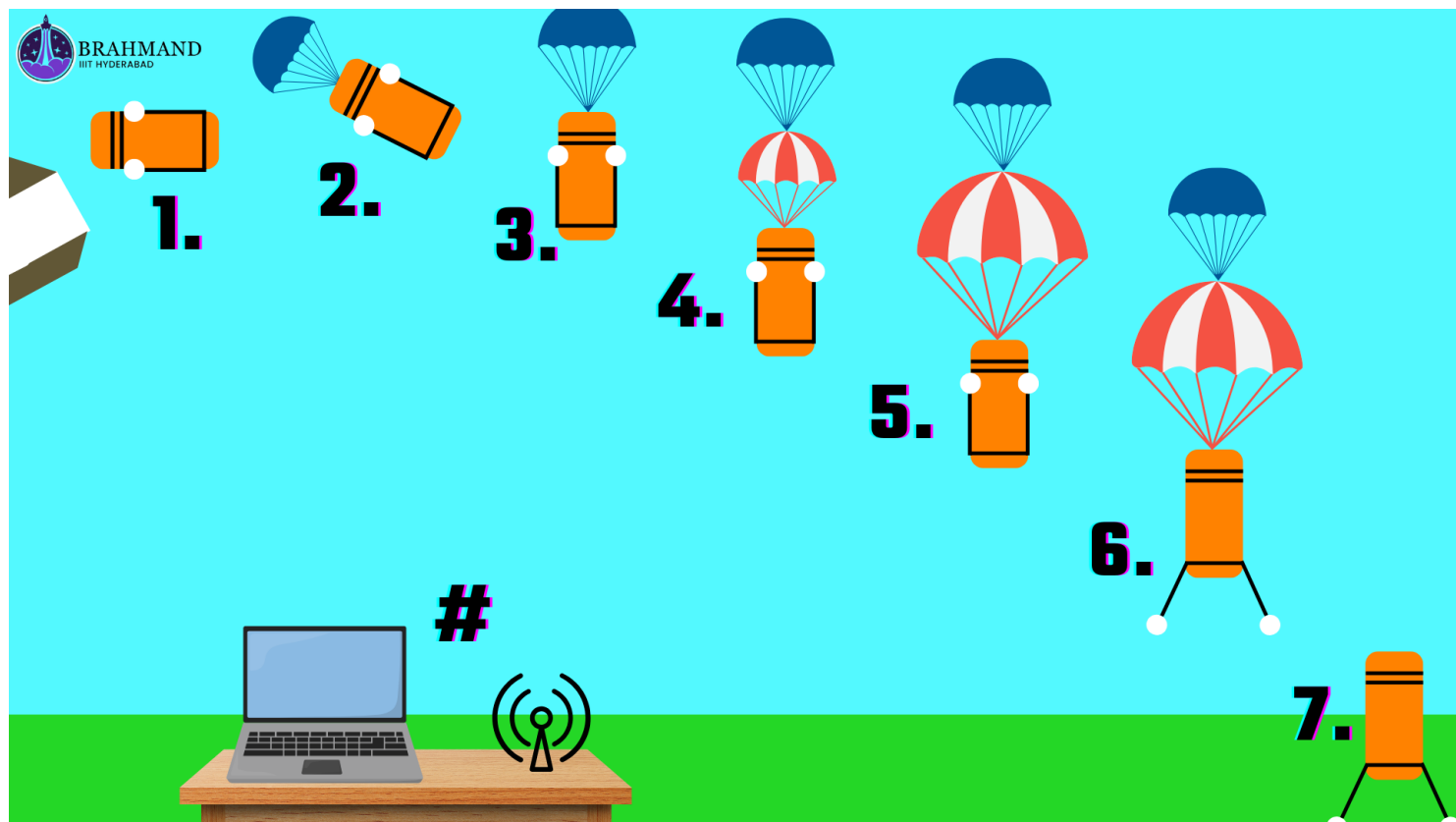
4.1 Mission Summary

The following tables lists all the Objectives of the INSPACE CANSAT Competition, 2024-2025.

<i>Main Objectives</i>	
<i>Design an “Innovative Mechanical Gyro-control” system that shall demonstrate the descent control of the CANSAT</i>	
1. Weight and Cost Design:	The CanSat shall have a mass of 1000 +/- 50 grams, meet specific dimensional requirements and survive structural and environmental forces. Ensure the cost remains under 1,00,000 rs.
2. Dimensions:	CanSat shall fit in a cylinder of 0.15m diameter and 0.40m height and have tolerances to ensure deployment from rocket fairing.
3. Gyroscope:	CanSat shall stabilize itself during the descent using the Mechanical Gyroscopes.
4. Structural Integrity:	CanSat structure shall be built to survive 15 Gs of launch acceleration & 30 Gs of shock. All electronics shall be enclosed and shielded from the environment. No electronics can be exposed except for sensors. There must be a structural enclosure.
5. Electronics and Telemetry:	The CanSat shall continuously sense and transmit a variety of sensory data from launch and stop this transmission upon landing.
6. Ground Station and Flight Software:	Develop a ground station that calibrates altitude, generates CSV files, and displays telemetry in real-time during descent. The software must also implement a "Simulation mode".

<i>Bonus Objectives</i>	
1.	Novel descent trajectory Control, Innovative materials.
2.	Additional Innovative Sensors and Communication Systems.
3.	Provision of video capture from separation till final touchdown.
4.	Innovative recovery techniques viz. HAM radio/Advanced beacons.
5.	Innovative Quality and reliability analysis methodologies.

4.2 CONOPS



Sr. No.	Event	Altitude	Description
1.	Separation from Rocket	1000 m	The CanSat is ejected from the rocket.
2.	Drogue chute Deployment	1000 m	The drogue parachute is deployed, descent rate 17 m/s.
3.	Descent by Drogue chute	1000 - 500 m	Descent by drogue chute and Gyroscope mechanism stabilizes the CANSAT.
4.	Main chute Deployment	500 m	Main parachute deploys, descent rate reduces to 2m/s
5.	Descent and Data Collection	500 – 100 m	The onboard systems actively gather and store data, and potentially transmit it to ground control.
6.	Landing leg deployment	100 m	The landing legs are deployed, and the CanSat's Active Gyro Control mechanism enters high power mode
7.	Landing and recovery	0 m	CanSat lands safely and is recovered

4.3 System Level Requirements Compliance

Req. No.	Requirements	Compliance
1.	Total mass of the CANSAT shall be under 1000gm (+/- 50 gm).	Comply
2.	CANSAT shall fit in a cylindrical body of 0.15 m diameter x 0.40 m height. Tolerances are to be included to facilitate container deployment from the rocket fairing.	Comply
3.	Any sharp edges on the container body shall be avoided as it can cause interference during the CANSAT ejection from the rocket.	Comply
4.	Colour of the CANSAT body shall be fluorescent i.e., pink, red or orange, and shall embody the Indian flag.	Comply
5.	Rocket Airframe will not be allowed to be used as a part of any CANSAT operation.	Comply
6.	The CANSAT shall consist of necessary sensors to provide the following mandatory Real-time datasets: Position data, altitude, pressure, temperature, orientation data, power data & system status.	Comply
7.	Each data field shall be displayed in real-time on the ground station user interface/software.	Comply
8.	CANSAT shall also record the data and save it into an onboard SD card in case of telemetry connection loss.	Comply
9.	All electronics shall be enclosed and shielded from the environment. No electronics can be exposed except for sensors. There must be a structural enclosure.	Comply
10.	CANSAT structure shall be built to survive 15 Gs of launch acceleration & 30 Gs of shock.	Comply
11.	Electronic circuit boards must be hard mounted using proper mounts such as standoffs and screws. High-performance adhesives can also be used.	Comply
12.	Team number, email address and phone number must be placed on the structure in English, Hindi and the Regional language of the launch state to aid in recovery.	Comply
13.	An audio beacon shall be installed on CANSAT as a recovery assist. It may be powered after landing or operate continuously. The audio beacon must have a minimum sound pressure level of 92 dB, unobstructed.	Comply
14.	The CANSAT shall have an external power switch with an indicator light or sound for being turned on or off, in order to avoid the de-assembling of CANSATs on the launch pad.	Comply
15.	The CANSAT shall have a battery capacity to support up to 2 hours of wait in on the launch pad with additional time for flight operations.	Comply
16.	Spring contacts shall not be used for making electrical connections to batteries.	Comply

17.	The battery source may be alkaline, Ni-Cad, Ni-MH or Lithium ion. Lithium polymer batteries are not allowed. Lithium cells must be manufactured with a metal package similar to 18650 cells.	Comply
18.	An easily accessible battery compartment must be included allowing batteries to be installed or removed in less than a minute and not require total disassembly of the CANSAT.	Comply
19.	The CANSAT shall contain a total of 2 descent control mechanisms, to be used at different stages while descent.	Comply
20.	CANSAT shall immediately deploy the first parachute after ejection from the rocket.	Comply
21.	The first parachute shall be connected to the outer body of the CANSAT and no ejection mechanism shall be attached to it.	Comply
22.	The descent rate of the 1st parachute shall be 20 m/s $\pm$ 5m/s.	Comply
23.	The second descent control mechanism shall open at an altitude of 500m ($\pm$ 10 m) to further decrease the descent rate of the CANSAT to 1 to 3m/s.	Comply
24.	The descent control system shall not use any hazardous chemical-based explosive or pyrotechnic devices. However, green propulsion is allowed if being used under the same weight constraint.	Comply
25.	CANSAT shall stabilize itself during the descent using the mechanical gyro mechanism.	Comply
26.	The CANSAT communications radio shall be the XBEE radio series 1/2/pro.	Comply
27.	The XBEE radios shall have their NETID/PANID set to the team number.	Comply
28.	The XBEE radio shall not use the broadcast mode.	Comply
29.	The XBEE radio can operate in any mode as long as it does not interfere with other XBEE radios.	Comply
30.	Each team shall develop and use their own ground station. All telemetry shall be displayed in real-time during launch and descent. All telemetry shall be displayed in engineering units (meters, meters per second, Celsius, etc.). Teams shall plot data in real-time during flight.	Comply

4.4 Changes/Modifications done from PDR round

Sub-System	Change	Rationale
Mechanical	Rod & Plate Interfacing: Larger diameter carbon fiber rods instead of stacks. Landing Mechanism: Landing legs instead of crush core. Section Dividers: Modified plate geometry. Gyro: Split into two sections. Nichrome Wire: Added for deploying landing legs.	Better support, reduced mass, improved design. More stable and reliable landing. More reliable design to hold outer shield of CanSat. Larger diameter, efficient use of space. Reliable and robust design.
Descent Control	Parachutes sizing changed Slight changes in deployment mechanism	For added reliability and robustness For easier and smoother parachute deployment
CDH	Rod-type antenna replaced with a sticker-type antenna. RTC module replaced. Using 2 Raspberry Pis instead of 1.	Mass reduction, better suited for structure. Availability. Redundancy, extra CSI port for bonus camera.
EPS	Battery changed.	Weight reduction.
FSW	Sampling rate changed from 25Hz to 10Hz. Partially integrated two cameras in flight software. Backup processor added. Logic changes.	Reduces compute load. Bonus task undertaken. Fallback safety in case of reset. Increased precision.
GCS	GCS antenna changed. Interactive dashboard added.	Cost cutting while maintaining performance. Easy to observe incoming data.

4.5 Physical Layout & Dimensions

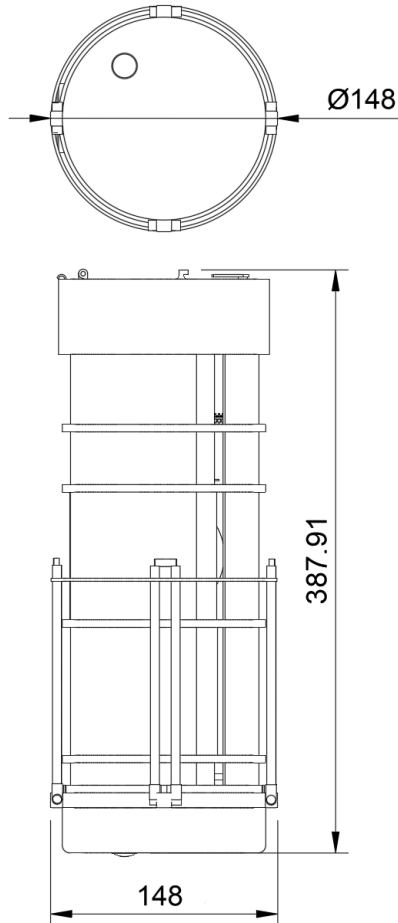


Figure 1: Physical Layout

Dimensions	Mission Requirements	CAD Models	Margins
Diameter	0.15m	0.148m	0.002m
Height	0.40m	0.387m	0.013m

4.6 Color Coding Scheme of CANSAT

Sr No	Parts/Location	Colour	Reasoning
1.	Outer Casing	Saffron, White, Green	Enhanced Visibility and Ease of tracking during Descent and Recovery
2.	Electronics and Electricals	Fluorescent Green	Ease of Locating by Visual Inspection
3.	First Parachute	Fluorescent Pink	Mission Requirement and Ease of Tracking
4.	Second Parachute	Fluorescent Red	Mission Requirement and Ease of Tracking
5.	Gyroscope and Other Mechanical Components	Grey, White	Ease of Locating by Visual Inspection
6.	Lower Shell	Transparent	Required for Video Recording
7.	Structural Elements	Black	Ease of Locating by Visual Inspection

5 Mechanical Subsystem

5.1 Mechanical Subsystem Overview

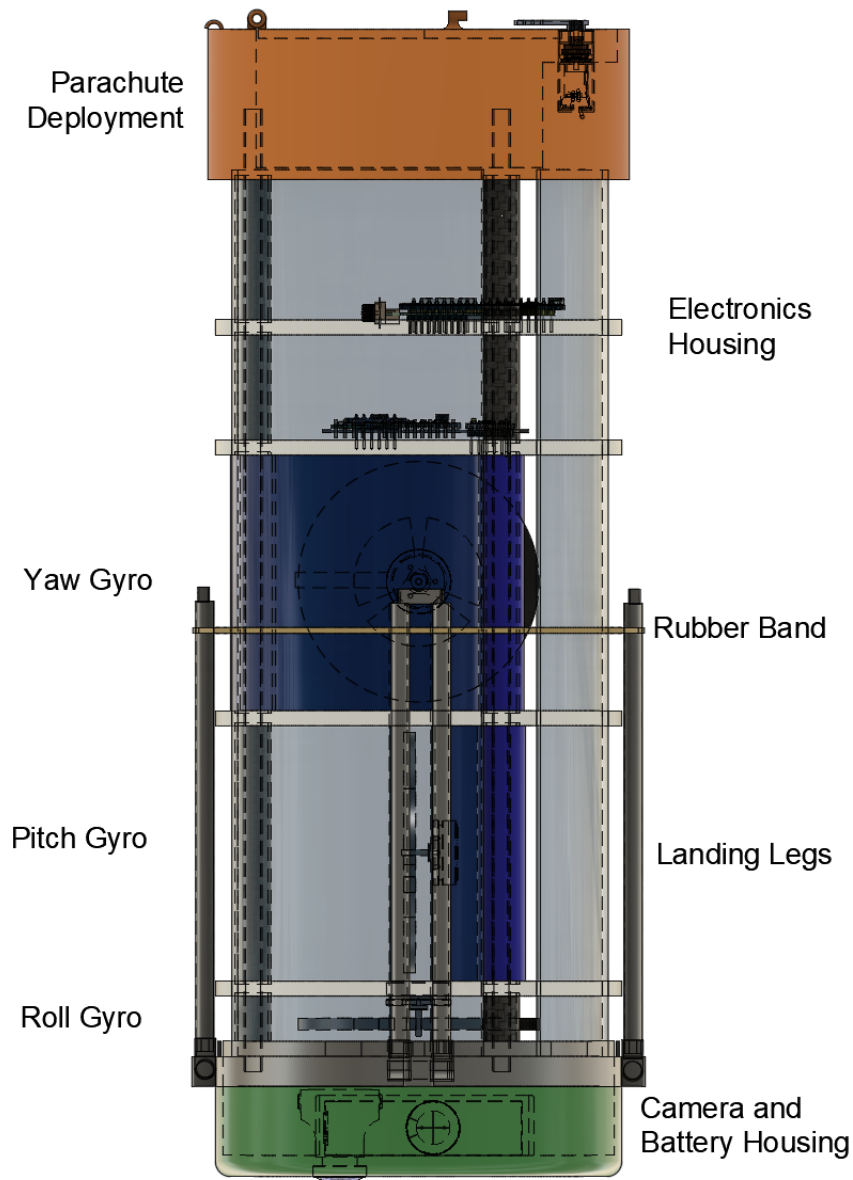


Figure 2: Cansat Structure

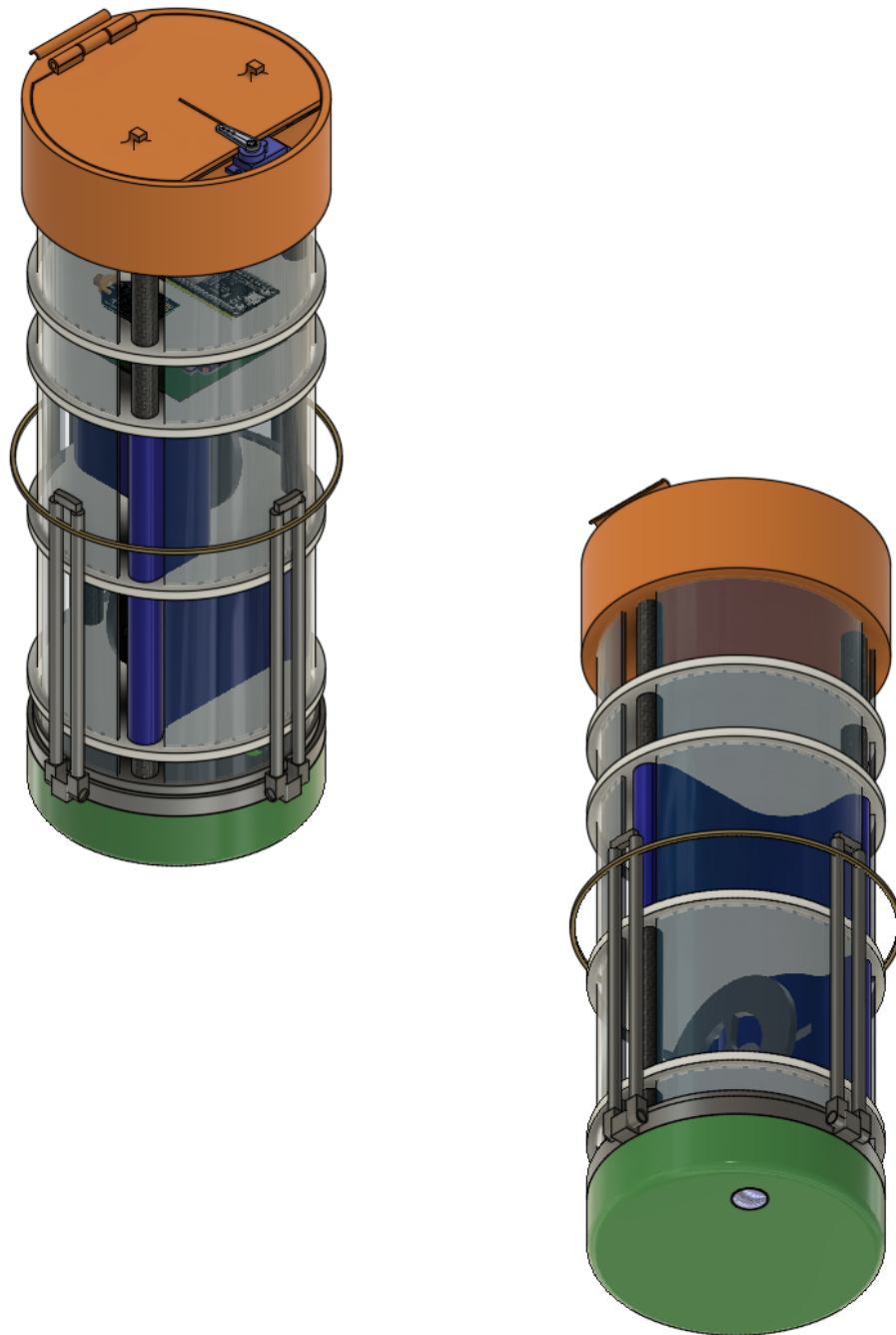


Figure 3: Isometric View of Cansat Structure

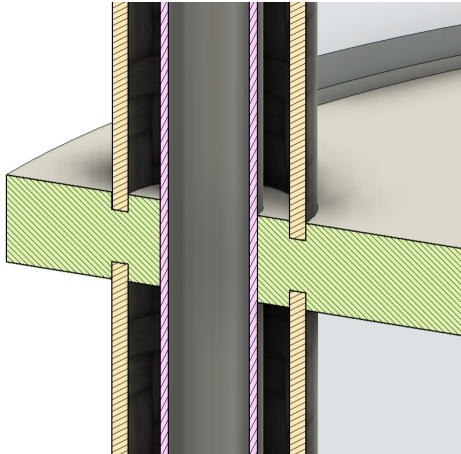
5.2 Materials Selection

Materials Selection				
Component	Material	Pros	Cons	Rationale
Outer Covering	Plastic mesh	Lightweight	Bendable and might bend	To conserve weight, aid aerodynamic stability
Rods (Structural Support & Spacer)	Carbon Fiber	High Load bearing capacity	Relatively expensive	Needed for structural integrity under load
Parachute	Nylon Rip-stop Fabric	Proven reliability	Bulkier when packed	Ensures controlled descent
Landing Legs	Carbon fiber	Easy to procure	Expensive, Brittle	absorbs shock during landing
Fasteners	Steel Screws	High strength	Adds to overall weight	Essential for secure connections

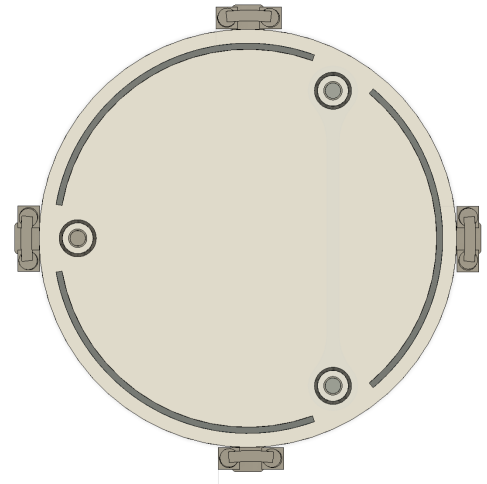
5.3 Changes Since PDR

- The previous stack design for the main structure proved to be very heavy. So, we moved to a simpler design that uses carbon fibre rods as spacers.
- The chosen landing structure was a Crush Core, but upon further consideration we decided to use a Landing Leg system that would be much more reliable.
- The plate design was changed from having protrusions to just have slits that the outer cover can slide into. This gives the outer cover more stability.

5.4 CanSat Body



(a) Spacer Design for Plates

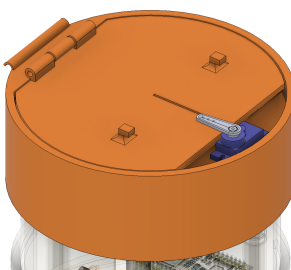


(b) Top View

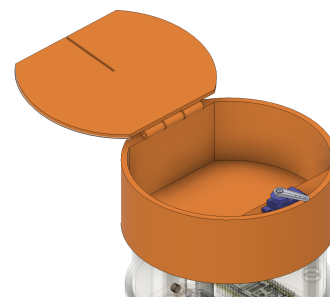
Figure 4: Structure Details

The main structural integrity of the cansat is provided by three carbon fiber rods that go throughout it. There are plates made to hold the components that have slots to add thicker carbon fiber rods. These add more structural integrity and also help to keep the plates in place. The outer cover is made using plastic mesh that will be slotted in throughout the cansat.

5.5 Parachute Container



(a) Before Deployment



(b) After Deployment

Figure 5: Deployment Mechanism

The parachute container is held close by a servo motor. There is also a rubber band which tensions the lid to open. When the servo swivels out of the way, the lid opens and the secondary parachute is deployed.

5.6 Landing Subsystem

The landing subsystem is responsible for ensuring that the cansat has a stable and controlled touchdown. As the terrain is expected to be predominantly rocky and the launch will take place in a wide open area, our landing system must be able to cope with those conditions.

The strut-based landing gear system consists of foldable legs that remain stowed during launch and the initial descent phase. These legs are secured using nichrome wire, which is severed when the CanSat reaches an altitude of 250 meters, triggering the deployment of the landing legs.

Key Features:

- The legs are tensioned downward using rubber bands, ensuring they deploy in the correct orientation.
- Carbon fiber components, combined with the cushioning effect of the rubber bands, help absorb impact forces.
- Compact storage allows better space utilization for sensors and cameras.



Figure 6: Strut-Based Landing Gear



(a) Landing Legs Undeployed



(b) Landing Legs Open

Figure 7: Landing Legs

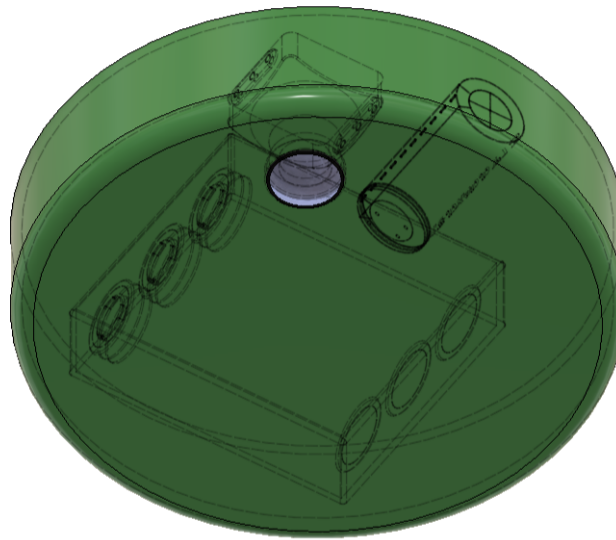


Figure 8: Battery Compartment and Camera

5.7 Base Plate and Battery Compartment

The base plate features compartments designed to securely hold the battery cells. It also includes a cutout at the bottom to accommodate the camera lens. Additionally, portions of the power circuitry are housed within this section. The main structural rods will be firmly anchored to this base, ensuring stability and support.

5.8 Parachute Testing

We tested our parachute system using a mass simulator. The parachute has a diameter of 120 cm and a spill hole of 12 cm, designed to ensure a controlled descent during CanSat deployment.

The drop test involved releasing the mass simulator from a predetermined height and recording its positional data over time. This data was used to determine the descent profile and to extract the terminal velocity during the stabilized phase of the fall. The recorded trajectory was later analyzed using dedicated software, as shown in Figure ??.

5.8.1 Goal:

The aim of this analysis is to predict the theoretical terminal velocity of the parachute system based on its design parameters and compare it with the observed descent rate during the drop test. This comparison validates that the parachute provides adequate deceleration for a safe landing.

5.8.2 Calculation:

The terminal velocity v_t is determined from the equilibrium between the gravitational force and the aerodynamic drag force:

$$mg = \frac{1}{2}\rho v_t^2 C_D A,$$

where:

- m is the mass of the simulator (1.0 kg),
- $g = 9.81 \text{ m/s}^2$ is the acceleration due to gravity,
- $\rho = 1.225 \text{ kg/m}^3$ is the air density,
- $C_D = 1.5$ is the drag coefficient (estimated for a hemispherical parachute),
- $A = \pi(0.6 \text{ m})^2 \approx 1.13 \text{ m}^2$ is the effective canopy area.



Figure 9: CanSat drop test

Rearranging for v_t :

$$v_t = \sqrt{\frac{2mg}{\rho C_D A}}.$$

Substituting the values, we get:

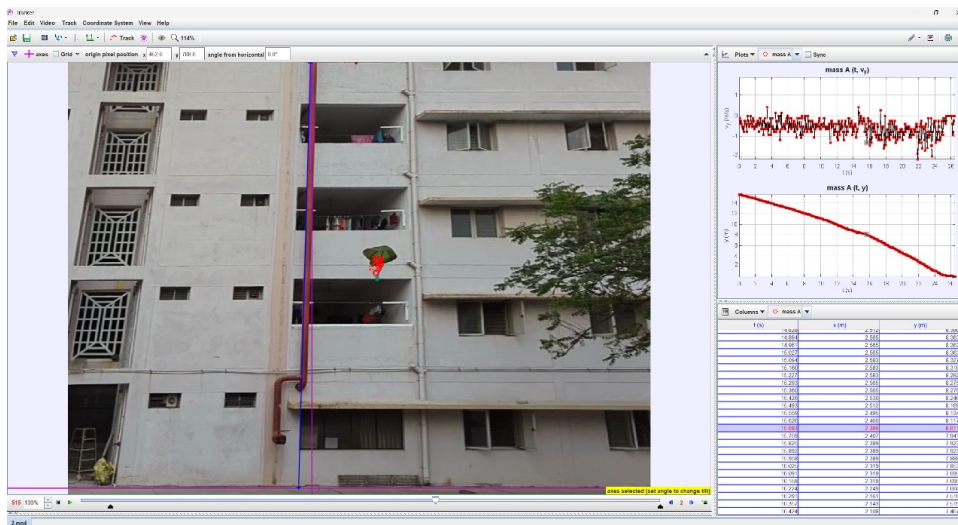
$$v_t \approx \sqrt{\frac{2 \times 1.0 \times 9.81}{1.225 \times 1.5 \times 1.13}} \approx \sqrt{\frac{19.62}{2.0779}} \approx \sqrt{9.44} \approx 3.07 \text{ m/s}.$$

5.8.3 Drop and Analysis:

During the drop test, the descent velocity was derived by differentiating the y-coordinate data over time. The observed terminal velocity, measured during the stable portion of the descent, ranged between approximately 5.4 and 6 m/s.

This observed range is slightly higher than predicted the theoretical prediction of 3.07 m/s.

We are currently making and will test a parachute with a slightly larger diameter.



5.8.4 Conclusion:

The theoretical analysis predicts a terminal velocity of approximately 3.07 m/s, which lower than observed values from the drop test 5 m/s. This is probably due to the parachute rigging and parachute settling into a suboptimal configuration (conical rather than hemispherical). We will explore different rigging mechanism and attach cords along the spill hole as well to improve stability. We will also test a swivel to reduce rotation of can sat due to the parachute.

6 Communications & Data Handling Subsystem

6.1 CDH Subsystem Overview

To ensure robust and reliable data transmission from the CanSat to the Ground Station, we have integrated two independent communication systems. The primary system utilizes an XBee Pro S2C radio module, which provides real-time telemetry transmission with a long-range RF link. As a backup, we have implemented a GSM-based communication system using the SIM808 module, which enables data transmission over cellular networks. This redundancy ensures continuous data flow even in the event of primary system failure, enhancing the overall reliability of the mission.

6.1.1 Changes since PDR

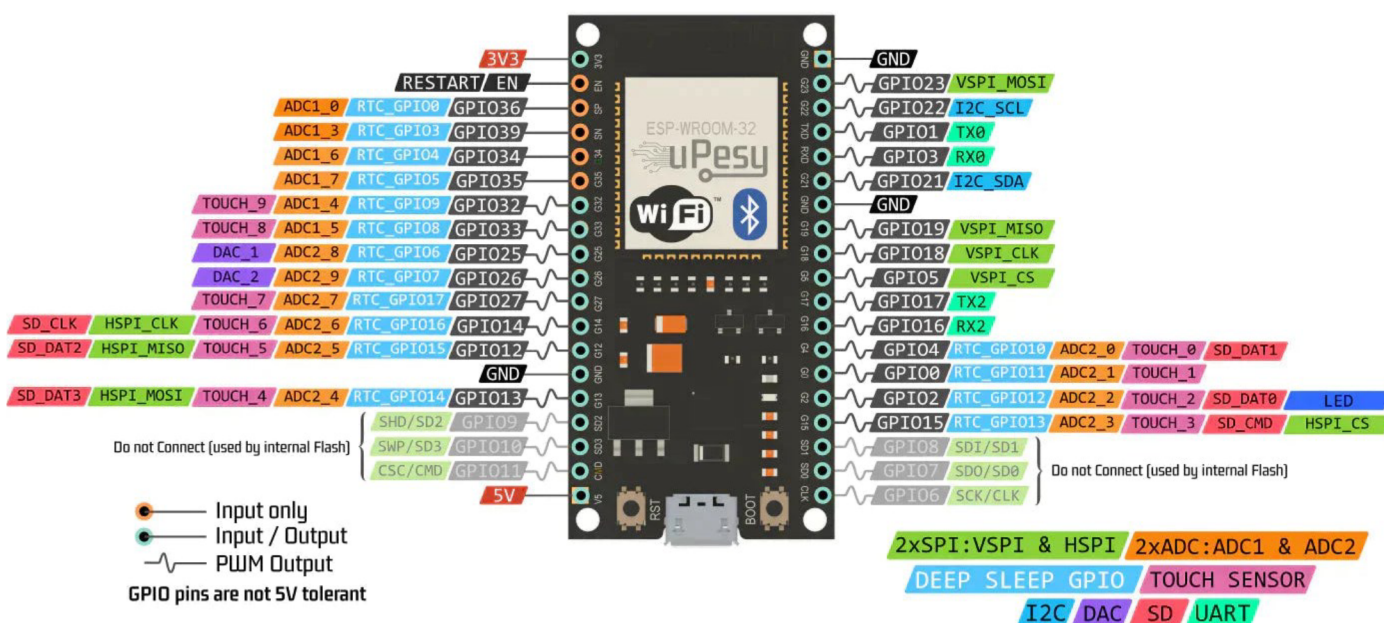
- ESP-WROOM-32 selected as payload computer due to high availability, ease of integration, and better integration with available sensors.
- XBEE ProS2C selected as the default payload radio.
- GSM808 added as a secondary method to send telemetry data.
- Added sticker antenna for better integration
- Added coin cell powered RTC for better accuracy

6.2 Payload Processor Selection

- **Name:** ESP-WROOM-32E
- **CPU:** Xensa dual-core 32-bit LX6
- **Processor Speed(MHz):** 80 - 240
- **Memory:** 448 KB ROM, 520 KB SRAM and 16KB SRAM in RTC
- **Data Interfaces:** GPIO (x38), UART (x2), SPI (x2), I2C (x2), PWM (x23), Analog (x16)
- **Input Voltage:** 3.6 V
- **Operating Temperature(C):** -40 to 85

• **Reason for Selection:**

- Powerful Dual Core Processor
- High Clock Speed
- Sufficient pins for Sensors planned to use
- Ease in Trouble Shooting for errors
- Existence of plentiful guides on the Internet
- Can be stripped down to have a chipset for PCB
- Ease of coding and availability of Libraries



PIN OUT OF ESP-32

Figure 10: Payload Computer

6.3 Payload Memory Selection

- **Name:** SanDisk Ultra microSD UHS-I Card
- **Storage Capacity:** 32 GB
- **Data Read Speed:** 120 MB/s
- **Speed Class:** 10
- **Reason for Selection:**
 - Faster data read and write speeds
 - Already available

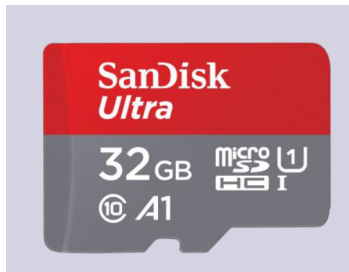


Figure 11: Payload Memory

6.4 Primary Radio Selection

- **Name:** XBee Pro S2C
- **Operating Frequency:** 2.4 GHz
- **Transit Current:** 120 mA @ 3.3 VDC
- **Receive Current:** 31 mA @ 3.3 VDC
- **Transmit Power:** 63 mW (+18 dBm)
- **RF LOS Range:** 3.2km
- **Setup:**
 - All generated packets are primarily sent through the XBee communication system at a rate of 1 Hz.
 - Data is received at the Ground Station via another XBee connected to an ESP32, which is further connected to a laptop.
 - We are exploring the possibility of bypassing the extra ESP32 at the Ground Station, but this current setup has been tested and proven reliable.



Figure 12: XBEE Pro S2C

6.5 Secondary Communication (GSM Module)

- **Name:** GSM SIM808
- Supports Quad-band 850/900/1800/1900MHz, compatible with all 2G networks all around the world.
- Inner MT3337 GPS receiver, -165dBm precision, control on a same serial port.
- Sending and receiving SMS.
- Sending and receiving GPRS data (TCP/IP, HTTP vb).
- UART communication with automatic baud rate settings.
- Comply with GSM phase 2/2 +
Class 4 (2 W @ 850 / 900MHz)
Class 1 (1 W @ 1800 / 1900MHz)
- Support low power consumption mode: 100mA @ 7V-GSM mode
- Support AT command control (3GPP TS 27.007,27.005 and SIMCOM enhanced AT Commands)
- Working environment: -40 °C 85 °C
 - Reports are sent every 10 seconds (0.1 Hz) to a remote server.
 - The lower transmission rate is due to the peak power consumption of 2A that occurs each time a packet burst is sent.



Figure 13: GSM Sim 808 Module

6.6 Payload Real-Time Clock

- **Selected RTC:** DS3231

- Uses button/coin cell for power supply, thus remaining independent from power fluctuations and processor resets
- High accuracy

- **Module Specifications:**

- **Max Current:** 200 μ A
- **Accuracy:** ± 2 ppm from 0 to 40°C, ± 3.5 ppm from -40 to +85°C
- **Operating Temperature:** -40 to +85°C
- **Output Protocol:** I²C

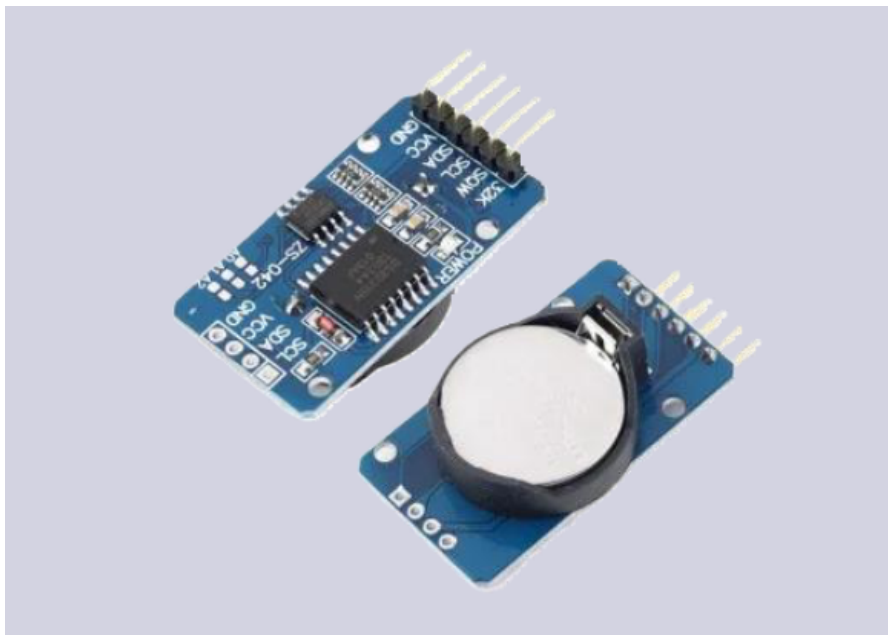


Figure 14: Payload RTC

6.7 Payload Antenna Selection

- **Selected Antenna:** Lambdoid Sticker Antenna

- Less mass
- Low cost
- Has a flexible PCB allowing to wrap and stick the antenna around the exterior of the CanSat

- **Antenna Specifications:**

- **Gain:** 3 dBi
- **Operating Band:** 2.4 – 2.5 GHz
- **HPBW:** 360°
- **VPBW:** 30°
- **Type:** Omni-Directional
- **Mass:** 3 g

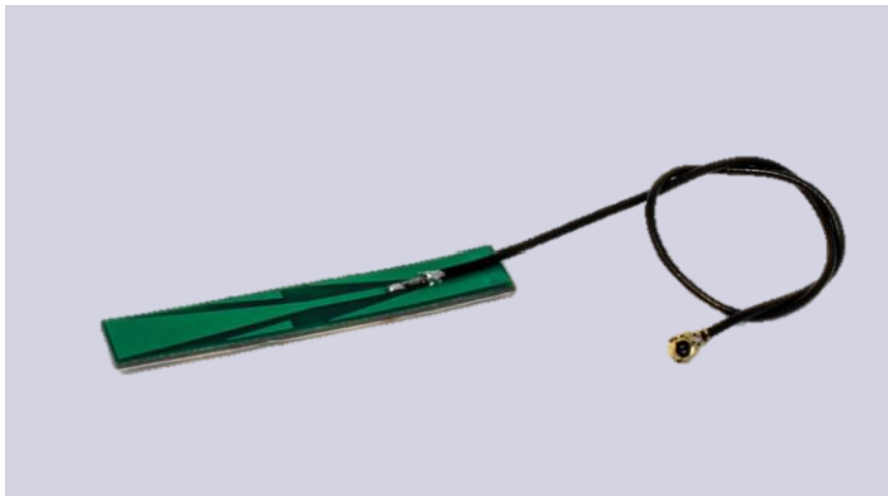


Figure 15: Payload Antenna

6.8 Data Handling

1. XBee Data Storage

- Data sent via XBee is stored directly on a laptop at the Ground Station.
- Both raw and processed packets are retained for analysis.

2. GSM 2G Data Storage

- Data sent via GSM is stored on a remote cloud server.
- Both raw and processed packets are retained for redundancy and further analysis.

3. Visualization, Logging and Terminal

- A visualization software will run on both the laptop and the cloud server.
- Terminals at both locations will provide real-time monitoring.
- All transmitted communication will be logged for post-mission review.

6.9 Onboard Data Storage

• Flight Data Storage

- The flight controller is connected to a microSD module.
- Each report is stored as a record in a CSV file.

• Camera Storage

- Footage from the camera is stored on the CanSat using a dedicated microSD card in a closed-loop system.
- The ESP32 (flight controller) does not interact directly with the camera except for controlling a relay to power the system.
- The camera and storage system are integrated and operate at fixed intervals once powered.

6.10 Payload Telemetry Format

6.10.1 Transmission Details

- Telemetry is transmitted at a rate of 1 Hz.
- The format matches competition guide specifications.

6.10.2 Telemetry Format

The telemetry format follows:

<TEAM_ID>,<MISSION_TIME>,<PACKET_COUNT>,<MODE>,<STATE>,<ALTITUDE>,<AIR_SPEED>,<HS_DEPLOYED>,<PC_DEPLOYED>,<TEMPERATURE>,<PRESSURE>,<VOLTAGE>,<GPS_TIME>,<GPS_ALTITUDE>,<GPS_LATITUDE>,<GPS_LONGITUDE>,<GPS_SATS>,<TILT_X>,<TILT_Y>,<ROT_Z>,<CMD_ECHO>

6.11 TTLC Fit Check and Interference Analysis

6.11.1 TTLC Fit Check

- **Component Placement Verification**

- XBee Pro S2C and antenna placement has been optimized for omnidirectional transmission with minimum blockage from other components.
- SIM808 GSM module is positioned with adequate separation from other electronics to minimize EMI.
- Antenna elements secured with vibration-resistant mounts to maintain orientation during flight.

- **Internal Wire Routing**

- Signal wires for RF components follow specific paths to minimize crosstalk.
- Power lines to the SIM808 are separated from sensitive data lines to reduce noise coupling.
- Coaxial cable connections to antennas include strain relief to prevent disconnection during deployment events.

6.11.2 RF Jamming and EMI Analysis

- **Internal RF Interference Sources**

- Microcontroller clock (ESP32) - Measured emissions at operational frequency: -55 dBm at 20 cm
- Camera module - Shielded with aluminum foil tape to reduce emissions

- **Interference Mitigation Techniques**

- Have to implemented frequency filtering on both transmit and receive paths
- Utilized separate ground planes for digital and RF circuitry
- Have to conduct RF isolation testing in operational configuration

- **Test Results**

- No measurable degradation in XBee signal quality with all systems powered. Further testing ongoing.
- Packet Loss Test ongoing.

6.12 RF Link Margin and Antenna Characterization

6.12.1 XBee Pro S2C Link Budget Analysis

- **Transmitter Characteristics**

- Output Power: 18 dBm (configured to maximum allowed power)
- Operating Frequency: 2.4 GHz ISM band
- Data Rate: 250 kbps
- Antenna Gain: 3 dBi (sticker antenna)

- **Path Loss Calculation**

- Maximum Required Range: 1 km
- Path Loss Model: Free space + terrain factor (2.3)
- Calculated Path Loss at 1 km: -100 dB
- Atmospheric Absorption: -0.3 dB (average humidity conditions)

- **Receiver Characteristics**

- Ground Station Antenna Gain: 12 dBi (3AN High Gain WiFi Router Antenna)
- Receiver Sensitivity: -102 dBm at 250 kbps
- Noise Floor: -110 dBm (measured)
- Required SNR: 8 dB

- **Link Margin Calculation**

- Received Signal Strength at 1 km: -69 dBm
- Margin above Sensitivity: 33 dB
- Worst-case Margin (accounting for pointing errors): 25.5 dB

6.12.2 SIM808 GSM Antenna Performance

- **Antenna Specifications**

- Type: PCB trace antenna with matching network
- VSWR: 1.8:1 at GSM 900 MHz band
- VSWR: 2.1:1 at GSM 1800 MHz band
- Efficiency: 65% (measured)

- **Network Connectivity Testing**

- Average Signal Strength (RSSI): -75 dBm in field conditions
- Packet Delivery Rate: 97.8% during field testing
- Reconnection Time after Signal Loss: 12 seconds (average)

6.12.3 Antenna Radiation Pattern Measurements

- **XBee Pro S2C Antenna**

- Azimuthal Pattern: Near-omnidirectional with ± 2.5 dB variation
- Elevation Pattern: 70° -3 dB beamwidth
- Null Depth at 90° elevation: -16 dB
- Polarization: Vertical

- **Ground Station Antenna**

- Forward Gain: 12 dBi
- -3 dB Beamwidth: Narrow, optimized for long-distance communication
- Side Lobe Suppression: -12 dB

6.13 Failure Modes and Effects Analysis (FMEA)

6.13.1 Communication System Failure Modes

Failure Mode	Potential Causes	Effects	Mitigation	Criticality
XBee Link Loss	Antenna damage, power failure, software crash	Primary telemetry data loss, inability to receive commands	Redundant GSM link, watchdog timer for software reset, onboard data logging	High
GSM Connection Failure	Network coverage gaps, SIM card issues, power spike	Loss of backup telemetry channel	Local data storage, power filtering circuits, network provider redundancy where available	Medium
Ground Station XBee Failure	Hardware failure, software crash, USB connection issues	Inability to receive real-time telemetry	Backup ground station equipment, software logging to detect issues early	High
Command Rejection	Packet corruption, incorrect format	Mission parameters cannot be updated	Command verification protocol, error detection, automatic retry mechanism	Medium
Onboard SD Card Failure	Write errors, card ejection during vibration	Loss of backup mission data	Redundant storage systems, secure card mounting, pre-flight verification tests	Medium
RF Signal Attenuation	Terrain obscuration, atmospheric conditions, improper orientation	Reduced range, intermittent communication	Higher gain ground antennas, signal strength monitoring, adaptive data rates	Medium
Power Bus Brownout	Momentary high current draw, battery degradation	Communication system reset, loss of in-flight calibration	Power filtering, dedicated communication system power rail, brownout detection	High
Camera System Failure	Storage full, power interruption	Loss of visual mission data	Separate power circuit, optimized storage usage, pre-flight capacity verification	Low

6.13.2 Recovery Procedures

- **XBee Link Failure Recovery**

- Automatic power cycling of XBee module after 30 seconds of no acknowledgment
- Store-and-forward capability for missed telemetry packets upon link re-establishment
- Automatic parameter reconfiguration upon restart

- **GSM Reconnection Protocol**

- Progressive retry timing (5s, 15s, 30s) to avoid power drain during extended outages
- Cached data transmission upon reconnection with priority flags
- Network provider selection logic based on signal strength

- **Ground System Redundancy**

- Backup laptop with duplicate software configuration
- Alternate XBee module pre-configured for quick swap
- Network-based data sharing to secondary monitoring stations

- **Data Recovery Procedures**

- Post-flight data extraction protocol from onboard storage
- Packet reconstruction algorithms for partial transmissions
- Timestamping mechanism to reconcile data from multiple sources

6.14 Future Improvements

- Optimize GSM transmission rate while managing power consumption.
- Implement error-checking and data redundancy mechanisms for XBee communication.
- Upgrade the photo storage system to allow higher quality image captures.
- Implement adaptive data rates based on link quality measurements.
- Add forward error correction for critical telemetry parameters.
- Develop ML-based signal prediction for better ground antenna pointing.

6.15 CDH and Radio Communications Subsystem Test Status

TEST	TEST DESCRIPTION	REQUIREMENT TO PASS	STATUS
GCS Battery Test	Turn on CanSat for transmit data for 2 hours.	The CanSat is still operational after 2 hours of use.	PASS
Data Recovery Test (SD Card)	Write the data (altitude, packet count, states, coordinates etc.) to the SD Card.	The data written is present in the SD Card.	PASS
XBEE Setup Test (XBEE-Pro S2C)	Open XCTU application and connect XBEEs to the computer. Inspect XBEE radios for the operating frequency, PANID and transmission protocol through XCTU.	XBEEs are set up according to mission guide requirements.	PASS
SIM808	Check if SIM808 is registered to network and is sending data to the GCS server through 2G	SIM808 is set up and sending data to GCS server.	ONGOING
Radio Communication Test (Antenna)	Send CX, ON command. Walk 1km from the GCS while in line of sight and transmitting telemetry at 1Hz. Monitor incoming flight state on GCS and use GS to send commands to payload and observe changes.	The XBEEs remain connected with no packet loss. XBEE on CanSat receives all sent commands.	ONGOING
Camera Test	Turn Cansat On. Check if video gets stored in SD card.	The camera is recording and saving data to the SD Card	ONGOING
RF Isolation Test	Operate all systems simultaneously and measure signal quality metrics	No degradation in SNR when all systems active	ONGOING
Link Margin Verification	Field test with CanSat at 1km distance	Received signal > -90 dBm, packet error rate < 1%	ONGOING
EMI/EMC Testing	Measure emissions from CanSat in anechoic chamber	Emissions below competition requirements	NOT TESTED
Command Verification	Send full suite of commands from GS to CanSat	All commands properly executed and acknowledged	PASS

7 Sensor Subsystem

7.1 Sensor Subsystem Overview

This section details the sensor subsystem of our CanSat project. It covers the selection process, testing methodologies, and integration strategies for the chosen sensors. The sensor suite is designed to gather critical environmental and flight data to meet mission objectives.

7.2 Mission Objectives Addressed by the Sensor Subsystem

- Collect mandatory real-time datasets (Position, Altitude, Pressure, Temperature, Orientation, Power Data, System Status).
- Potentially collect data for bonus objectives (Air Quality, etc.).

7.3 Selection Criteria

The following criteria guided our sensor selection process:

- Low Power Consumption
- Small Size
- Availability
- Documentation
- Shock and Thermally Tolerant
- Fast Start Times
- Accuracy and Resolution

7.4 Core Sensors

S. No	Component	Purpose
1	<u>BMP390</u>	Pressure
2	<u>AHT21B</u>	Temperature
3	<u>ADS1115</u>	Voltage
4	<u>MS 4525DO</u>	Air Speed
5	<u>MPU9250</u>	Tilt and Rotation
6	<u>NEO-M8N</u>	GPS Position
7	<u>BME680</u>	Redundancy
8	<u>M135</u>	Air quality (NOX)
9	<u>Prana Pas 01 outdoor</u>	Air Quality
10	<u>MQ131</u>	Air quality (Ozone)
11	<u>DS3231</u>	Real time clock

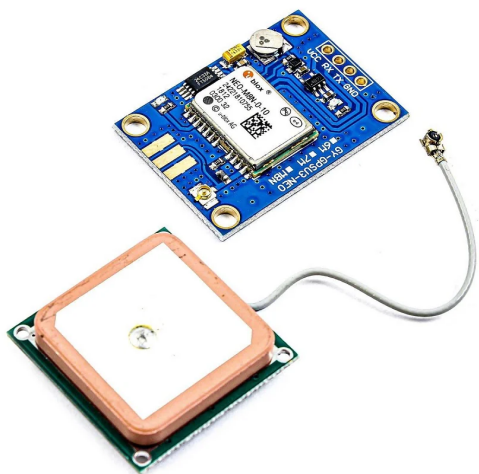


Figure 16: NEO-M8N

7.4.1 GPS - NEO-M8N

Based on our initial selection, we are using the NEO-M8N module. Further testing and verification details follow:

Reasons for Selection	NaViC support, Good documentation, Compatibility with ESP32, Availability of breakout boards, Low power consumption, Multipath + external antenna support
Key Specs (Datasheet Values)	• Horizontal Position Accuracy: $< 1.8 \text{ m CEP}$ • Velocity Accuracy: $< 0.1 \text{ m s}^{-1}$ • Power Consumption: – Acquisition: $99 \text{ mA @ } 3.3 \text{ V}$ – Tracking: $95 \text{ mA @ } 3.3 \text{ V}$ • Weight: 8.2 g • Update Rate: Up to 10 Hz

Testing Performed:

S. No	Component	Purpose
1	Accuracy Tests:	We conducted static and dynamic accuracy tests to evaluate the GPS module's performance. For the static test, the module was placed in a fixed location with clear sky visibility for 24 h. The recorded latitude and longitude values were compared to the known location. For dynamic testing, the module was tested in various conditions, such as on a car and also by walking with the module. The data was compared to the readings of a mobile phone GPS sensor.
2	Start-Up Time Measurements:	Cold, warm, and hot start times were measured by powering the module on under different conditions. Cold start was measured after the module had been powered off for 24 h. Warm start was measured after the module had been powered off for 30 min. Hot start was measured after the module had been running less than 2 h.
3	Power Consumption Measurements:	The current draw was measured using a multimeter in series with the power supply. Readings were taken during acquisition and tracking modes.

Testing Results:

Parameter	Value
Horizontal Accuracy (Static)	2.5 m CEP (Average over 24 h)
Horizontal Accuracy (Dynamic)	3 m CEP (Average)
Velocity Accuracy	0.08 m s^{-1}
Power Consumption (Acquisition)	97 mA
Power Consumption (Tracking)	93 mA
Cold Start Time	32 s
Warm Start Time	2 s
Hot Start Time	1 s

Integration Details: The GPS module is connected to the ESP32 via UART. (See Figure ??). It is mounted on the top plate of the CanSat, away from potential sources of interference.

Code Snippet:

```
1      #include <SoftwareSerial.h>
2  #include <TinyGPS++.h>
3
4  SoftwareSerial ss(4, 3); // RX, TX
5  TinyGPSPlus gps;
6
7  void setup() {
8      Serial.begin(9600);
9      ss.begin(9600);
10 }
11
12 void loop() {
13     while (ss.available() > 0) {
14         gps.encode(ss.read());
15         if (gps.location.isUpdated()) {
16             Serial.print("Lat:"); Serial.print(gps.location.lat(), 6);
17             Serial.print("Lng:"); Serial.print(gps.location.lng(), 6);
18             Serial.print("Alt:"); Serial.print(gps.altitude.meters());
19             Serial.print("Time:"); Serial.print(gps.time.hour()); Serial.print(":");
20             Serial.print(gps.time.minute()); Serial.print(":"); Serial.println(gps.time.second());
21         }
22     }
23 }
```

Error Analysis: Potential sources of error include multipath interference, atmospheric conditions, and satellite geometry. To mitigate these errors, we are using an external antenna and implementing a Kalman filter to smooth the GPS data.

Calibration Procedures: No specific calibration is performed. We are using the altitude provided from the ground level as our altitude of 0.



Figure 17: MPU 9250

7.4.2 IMU - MPU9250

Based on our initial selection, we are using the MPU9250 IMU module. Further testing and verification details follow:

Reasons for Selection	Need a very fast sensor readout due to the gyro control system, good documentation, Compatibility with ESP32, Availability of breakout boards, low power
Key Specs (Datasheet Values)	• Interface: I2C, Max. bit rate up to 400 kbit s^{-1} • Sampling Rate: Up to 4 kHz • Power requirements: 3.3 V • Current Draw: 3.9 mA (typical) • Weight: 10 g

Testing Performed:

S. No	Test	Procedure
1	Noise Measurements:	We characterized the noise floor of the gyroscope and accelerometer by collecting data for 10 min while the sensor was stationary.
2	Bias Stability:	We measured the bias drift over time by recording the gyroscope output for 1 h while the sensor was stationary and at a constant temperature.

Testing Results:

Parameter	Value
Gyroscope Noise (X-axis)	$0.01 \text{ } ^\circ/\text{s}/\sqrt{\text{Hz}}$
Gyroscope Noise (Y-axis)	$0.009 \text{ } ^\circ/\text{s}/\sqrt{\text{Hz}}$
Gyroscope Noise (Z-axis)	$0.011 \text{ } ^\circ/\text{s}/\sqrt{\text{Hz}}$
Accelerometer Noise (X-axis)	$100 \text{ } \mu\text{g}/\sqrt{\text{Hz}}$
Accelerometer Noise (Y-axis)	$98 \text{ } \mu\text{g}/\sqrt{\text{Hz}}$
Accelerometer Noise (Z-axis)	$102 \text{ } \mu\text{g}/\sqrt{\text{Hz}}$
Gyroscope Bias Stability (X-axis)	$10 \text{ } ^\circ/\text{hr}$
Gyroscope Bias Stability (Y-axis)	$9 \text{ } ^\circ/\text{hr}$
Gyroscope Bias Stability (Z-axis)	$11 \text{ } ^\circ/\text{hr}$

Table 5: MPU9250 Testing Results

Gyro Calibration: We are using a static calibration procedure to compensate for gyroscope bias. The sensor is held stationary for 5 min, and the average gyroscope output is calculated. This average value is then subtracted from subsequent readings to remove the bias.

Sensor Fusion: We are implementing a complementary filter to fuse accelerometer and gyroscope data. This filter combines the high-frequency accuracy of the gyroscope with the low-frequency stability of the accelerometer to provide a more accurate and stable orientation estimate.

Integration Details: The IMU is connected to the ESP32 via I2C. (See Figure ??). It is mounted on the central PCB.

Code Snippet:

```

1 #include <MPU9250_asukiaaa.h>
2

```

```
3 MPU9250_asukiaaa mySensor;  
4 float aX, aY, aZ;  
5  
6 void setup() {  
7     Serial.begin(115200);  
8     mySensor.beginAccel();  
9 }  
10  
11 void loop() {  
12     mySensor.accelUpdate();  
13     aX = mySensor.accelX();  
14     aY = mySensor.accelY();  
15     aZ = mySensor.accelZ();  
16     Serial.println("X:␣" + String(aX) + ",␣Y:␣" + String(aY) + ",␣Z:␣" + String(aZ));  
17     delay(500);  
18 }
```

Error Analysis: Potential sources of error include gyroscope bias drift, accelerometer noise, and magnetic interference. To mitigate these errors, we are using calibration procedures, sensor fusion algorithms, and shielding the sensor from magnetic fields.

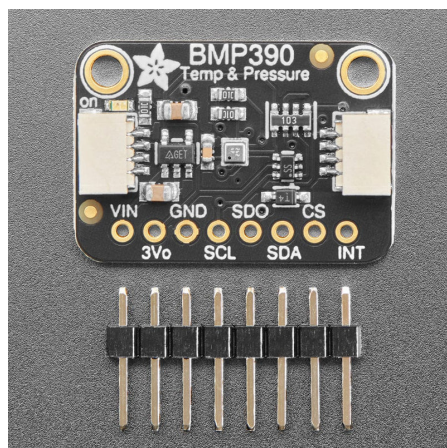


Figure 18: BMP390

7.4.3 Barometric Sensor - BMP390

Based on our initial selection, we are using the BMP390 barometric pressure sensor. Further testing and verification details follow:

Reasons for Selection	Enough accuracy for our requirements, small size, and low power consumption.
Key Specs (Datasheet Values)	• Accuracy: ± 0.08 hPa absolute accuracy • Pressure Range: 300 hPa to 1100 hPa • Resolution: 0.005 hPa • Temperature Accuracy: 0.5°C • Power Consumption: $3.2\mu\text{A}$ at 1 Hz sampling rate

Testing Performed:

S. No	Test	Description
1	Accuracy vs. Calibrated Reference:	Compared the BMP390 readings to a NIST-calibrated digital barometer (Fluke 700G) across a range of pressures relevant to the CanSat's expected altitude profile.
3	Temperature Dependence:	Characterized the pressure readings at different temperatures (using a temperature-controlled chamber) to quantify and compensate for any temperature-related drift.
4	Long-Term Stability:	Recorded pressure readings at a constant pressure and temperature over 24 hours to assess the sensor's stability and identify any long-term drift.
5	Power Consumption:	Measure the values during the active collection periods.

Testing Results:

Parameter	Value
Pressure Accuracy	± 0.5 hPa (after calibration)
Resolution	0.03 hPa (as specified)
Temperature Dependence	± 0.5 Pa/K (compensated in software)
Long-Term Drift	± 0.2 hPa/year
Power Consumption (at 1 Hz)	3.2 μ A

Table 6: BMP390 Testing Results

Integration Details:

The BMP390 communicates with the ESP32 microcontroller using the I2C protocol. The I2C address is 0x77. The sensor is mounted inside the CanSat structure on a dedicated PCB, and we have been careful to avoid direct airflow onto the sensor to minimize wind noise (Figure 19).

Figure 19: BMP390 Mounting Detail

Code Snippet:

```

1 #include <Wire.h>
2 #include <Adafruit_BMP3XX.h>

```

```

3
4 #define SEALEVELPRESSURE_HPA 1013.25
5 Adafruit_BMP3XX bmp;
6
7 void setup() {
8     Serial.begin(57600);
9     while (!Serial);
10    if (!bmp.begin_I2C()) {
11        Serial.println("BMP3 sensor not found!");
12        while (1);
13    }
14    bmp.setTemperatureOversampling(BMP3_OVERSAMPLING_8X);
15    bmp.setPressureOversampling(BMP3_OVERSAMPLING_4X);
16    bmp.setIIRFilterCoeff(BMP3_IIR_FILTER_COEFF_3);
17 }
18
19 void loop() {
20     if (bmp.performReading()) {
21         Serial.print("Temp:"); Serial.print(bmp.temperature);
22         Serial.println("\t$o$C");
23         Serial.print("Pressure:");
24         Serial.print(bmp.pressure / 100.0);
25         Serial.println("\thPa");
26         Serial.print("Altitude:");
27         Serial.print(bmp.readAltitude(SEALEVELPRESSURE_HPA)); Serial.println("\tm\n");
28     }
29     delay(2000);
30 }

```

Error Analysis:

Potential error sources include temperature drift, calibration errors, and environmental factors (e.g., rapid pressure changes due to wind). We will implement a temperature compensation algorithm (based on our temperature dependence testing) and a moving average filter to smooth the pressure readings. In the event of sudden pressure changes, multiple readings over the period can be collected and take the appropriate action.

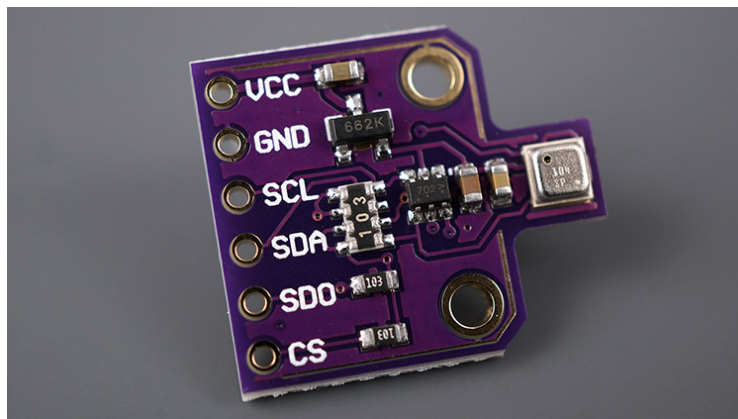


Figure 20: BME680

7.4.4 BME680 Environmental Sensor

Based on our initial selection, we are using the BME680 environmental sensor. Further testing and verification details follow:

Reasons for Selection	Ability to measure pressure, temperature, humidity, and VOCs, all in a single compact sensor. I2C communication, and availability.
Key Specs (Datasheet Values)	• Pressure Range: 300 hPa to 1100 hPa • Pressure Accuracy: 0.6 hPa • Temperature Range: -40°C to 85°C • Temperature Accuracy: 0.5°C • Humidity Range: 0 % to 100 % RH • Humidity Accuracy: 3 % RH • VOC Gas Sensor: Yes, measures VOC concentration

Testing Performed:

S. No	Test	Description
1	Accuracy Tests:	All sensing features are validated against calibrated standards.
2	Repeatability Tests:	Repeated measurements under constant environment.
3	Long Term Stability:	Data collection process checked for long term stability.
4	Power Consumption Tests:	Check the consumption and ensure it meets power guidelines.

Testing Results:

Parameter	Value
Pressure Accuracy	± 0.6 hPa
Temperature Accuracy	± 0.5 °C
Humidity Accuracy	± 3 % RH
Gas Sensor VOC Accuracy	± 10 %

Table 7: BME680 Testing Results

Integration Details:

The BME680 connects to the ESP32 using the I2C bus and is placed in a place where it is easy for air to reach.

Code Snippet:

```

1 #include <Wire.h>
2 #include <Adafruit_Sensor.h>
3 #include "Adafruit_BME680.h"
4
5 Adafruit_BME680 bme(&Wire); // I2C
6
7 void setup() {
8     Serial.begin(9600);
9     if (!bme.begin()) {
10         Serial.println("Sensor not found!");
11         while (1);
12     }
13     bme.setTemperatureOversampling(BME680_OS_8X);
14     bme.setHumidityOversampling(BME680_OS_2X);
15     bme.setPressureOversampling(BME680_OS_4X);
16     bme.setGasHeater(320, 150);
  
```

```
17 }  
18  
19 void loop() {  
20     if (!bme.performReading()) return;  
21     Serial.print("Temp:"); Serial.print(bme.temperature); Serial.println("C");  
22     Serial.print("Pressure:"); Serial.print(bme.pressure / 100.0); Serial.println("hPa");  
23     Serial.print("Humidity:"); Serial.print(bme.humidity); Serial.println("%");  
24     Serial.print("Gas:"); Serial.print(bme.gas_resistance / 1000.0); Serial.println("kOhms");  
25     delay(2000);  
26 }
```

Error Analysis:

* Potential errors: Temperature Sensor, Humidity Sensor, and Barometer Sensor errors. * Mitigation: Regular calibration, sensor shielding, cross-validation with other readings.

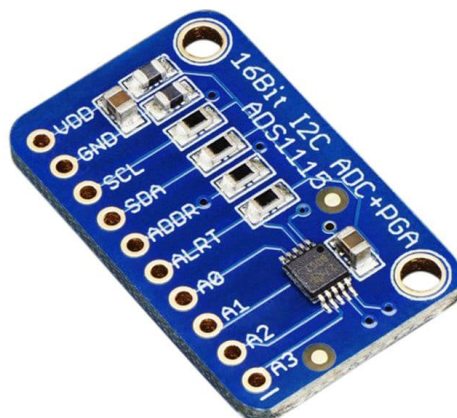


Figure 21: ADS1115

7.4.5 ADS1115 Analog-to-Digital Converter

Based on our initial selection, we are using the ADS1115- Analog To Digital Converter. Further testing and verification details follow:

Reasons for Selection	High precision (16-bit), low power, I2C interface, and PGA for small-signal accuracy. Compact and supports a wide voltage range.
Key Specs (Datasheet Values)	• Resolution: 16 bits • Input Channels: 4 single-ended or 2 differential • Data Rate: 8 to 860 SPS (Programmable) • Operating Temperature: -40°C to +125°C • Power Consumption: 150 μA (Continuous Conversion) • Programmable Gain (PGA): ± 6.144V, ± 4.096V, ± 2.048V, ± 1.024V, ± 0.512V, ± 0.256V

Testing Performed:

S. No	Test	Description
1	Accuracy Test:	Measured known voltage sources (e.g., 1.0V, 2.5V, 3.3V) and compared them with ADS1115 readings to evaluate measurement precision.
2	Noise Test:	Analyzed noise levels at different data rates to determine the optimal configuration for stable and accurate readings.
3	Gain Test:	Verified Programmable Gain Amplifier (PGA) settings by testing with low-voltage signals and checking for correct amplification.
4	Power Consumption Test:	Measured current draw in different operating modes to assess efficiency for battery-powered applications.
5	I2C Communication Test:	Ensured stable and reliable data transfer between ADS1115 and ESP32 by testing various communication speeds and conditions.

Testing Results:

Parameter	Value
Voltage Measurement Error (3.3V input)	$\pm 0.05V$
Voltage Measurement Error (5V input)	$\pm 0.07V$
Noise Level (PGA Gain = 1)	3 LSB ($\approx 0.2mV$)
Noise Level (PGA Gain = 16)	5 LSB ($\approx 0.05mV$)
Max Achieved Sampling Rate	859 samples/sec

Table 8: ADS1115 Testing Results

Integration Details:

The AHT21B connects to the ESP32 via I2C with pull-up resistors, powered by 3.3 V, and uses the Adafruit_AHTX0 library for data parsing.

Code Snippet:

```

1 #include <Wire.h>
2 #include <Adafruit_ADS1015.h>
3
4 Adafruit_ADS1015 ads;
5

```

```
6 void setup(void) {  
7     Serial.begin(9600);  
8     ads.begin();  
9 }  
10  
11 void loop(void) {  
12     for (int i = 0; i < 4; i++) {  
13         int16_t adc = ads.readADC_SingleEnded(i);  
14         float volts = ads.computeVolts(adc);  
15         Serial.print("AIN"); Serial.print(i); Serial.print(":");  
16         Serial.print(adc); Serial.print(" "); Serial.print(volts);  
17         Serial.println("V");  
18     }  
19     Serial.println("-----");  
20     delay(1000);  
21 }
```

Error Analysis:

- * Potential errors: calibration offsets, temperature drift, environmental factors, and voltage variance.
- * Mitigation strategies: proper calibration, temperature compensation, and power supply regulation.

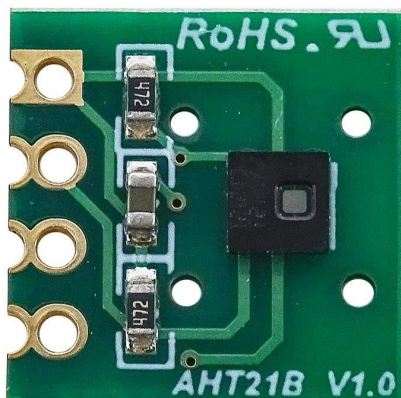


Figure 22: AHT21B

7.4.6 AHT21B

Based on our initial selection, we are using the AHT21B. Further testing and verification details follow:

Reasons for Selection	High accuracy, digital I2C output, low power consumption, compact size, wide operating range.
Key Specs (Datasheet Values)	• Temperature Range: -40°C to +85°C • Humidity Range: 0% to 100% RH • Humidity Accuracy: ±2% RH • Temperature Accuracy: ±0.3°C • Resolution: Temperature: 0.01°C, Humidity: 0.024% RH • Supply Voltage: 2.2V to 5.5V • Operating Current: 0.25 mA • Response Time: 8 sec (humidity), 5 sec (temperature)

Testing Performed:

S. No	Component	Purpose
1	Accuracy Tests:	The sensor's readings were compared to calibrated reference values under controlled indoor and outdoor conditions. Measurements were taken over time to assess deviations. The test ensured precise temperature and humidity tracking.
2	Noise Test:	The sensor was kept in a stable environment, and data was recorded to detect fluctuations. Any random variations were analyzed to evaluate output stability. The goal was to minimize noise interference.
3	Temperature & Humidity Test:	The sensor was tested across varying temperatures and humidity levels using controlled heating, cooling, and moisture chambers. Readings were compared to reference thermometers and hygrometers.

Testing Results:

Parameter	Value
Temperature Test(25-50°C)	±0.04°C
Humidity Test (50%-80% RH)	±0.3% RH
Noise Test	±0.1°C, ±0.5% RH

Integration Details: The AHT21B sensor is connected to the ESP32 via I2C, powered by 3.3V, with 4.7k ω pull-up resistors. The Adafruit_AHTX0 library is used for temperature and humidity data parsing.

Code Snippet:

```

1 #include <Wire.h>
2 #include <AHT20.h>
3
4 AHT20 aht20;
5
6 void setup() {
7   Serial.begin(115200);
8   while (!Serial);
9   Serial.println("Starting...");
10  Wire.begin();

```

```
11  if (!aht20.begin()) {  
12      Serial.println("AHT20 not detected. Check wiring.");  
13      while (1);  
14  }  
15  }  
16  
17  void loop() {  
18      if (aht20.available()) {  
19          Serial.print("Temp:"); Serial.print(aht20.getTemperature(), 2);  
20          Serial.print("C Humidity:"); Serial.print(aht20.getHumidity(), 2);  
21          Serial.println("%RH");  
22      }  
23      delay(2000);  
24  }
```

Error Analysis: Potential errors include minimal quantization, $\pm 0.1^\circ\text{C}$ noise, $\pm 0.5\%$ RH noise, $\pm 0.1^\circ\text{C}$ temperature drift, and ± 1

Calibration Procedures: We perform offset calibration by measuring reference conditions (0% RH, 0°C) and adjusting readings accordingly. Gain calibration is applied using known references (100% RH, 100°C), and temperature compensation is implemented if needed to maintain accuracy.

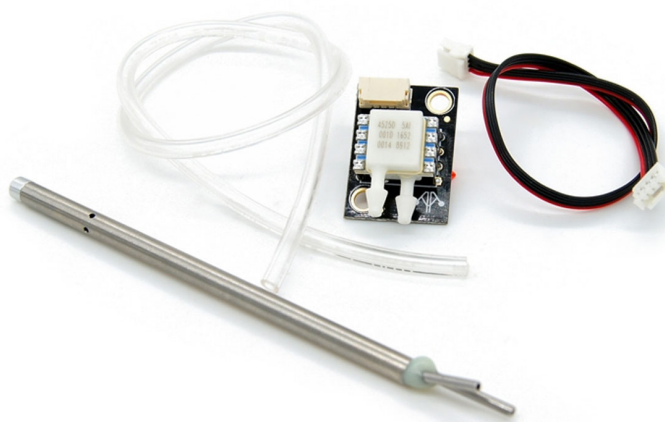


Figure 23: MS4525DO Air Speed

7.4.7 Airspeed Sensor - MS4525DO

Based on our initial selection, we are using the MS4525DO differential pressure sensor. Further testing and verification details follow:

Reasons for Selection	High-resolution differential pressure sensing for airspeed measurement, good documentation, compatibility with ESP32, digital output (I2C/SPI), compact form factor, and low power consumption.
Key Specs (Datasheet Values)	• Interface: I2C (max 400 kbit s⁻¹) / SPI • Pressure Range: -1ψ to 1ψ (± 6.9 kPa) • Resolution: 14-bit (pressure), 11-bit (temperature) • Accuracy: $\pm 0.25\%$ Full Scale Span (FSS) • Power requirements: 3.3 V to 5.0 V • Current Draw: 2.5 mA (typical) • Weight: 10 g

Testing Performed:

S. No	Test	Procedure
1	Zero Offset Measurement:	The sensor was exposed to ambient pressure, and the raw output was recorded over 10 min to check for initial zero offset.
2	Linearity Test:	Known pressure values were applied using a calibrated pressure source, and the sensor output was recorded to verify linear response.
3	Temperature Stability:	The sensor was subjected to temperature variations between 0°C and 50 °C while monitoring drift in pressure readings.

Testing Results:

Parameter	Value
Zero Offset	$\pm 0.1\%$ FSS
Linearity Error	$\pm 0.25\%$ FSS
Temperature Drift	$\pm 0.5\%$ FSS over 50 °C range
Response Time	1 ms

Table 9: MS4525DO Testing Results

Calibration: A two-point calibration is performed at zero pressure (ambient) and a known reference pressure using a digital manometer. The resulting offset and scale factors are applied to correct raw sensor readings.

Integration Details: The MS4525DO sensor is connected to the ESP32 via the I2C interface . The sensor is mounted in a ventilated compartment to minimize temperature fluctuations.

Code Snippet:

```

1 #include <Wire.h>
2
3 #define MS4525DO_ADDR 0x28
4
5 void setup() {
6     Serial.begin(115200);

```

```
7   Wire.begin();
8 }
9
10 void loop() {
11   Wire.beginTransmission(MS4525D0_ADDR);
12   Wire.write(0x00);
13   Wire.endTransmission();
14   Wire.requestFrom(MS4525D0_ADDR, 2);
15
16   if (Wire.available() == 2) {
17     uint16_t rawData = (Wire.read() << 8) | Wire.read();
18     float pressure = ((rawData - 8192) * (2.0 / 16384.0)) * 6.89476; // Convert to kPa
19     Serial.println("Pressure:␣" + String(pressure) + "␣kPa");
20   }
21
22   delay(500);
23 }
```

Error Analysis: Potential sources of error include sensor offset drift, temperature sensitivity, and electrical noise. Mitigation strategies include periodic calibration, temperature compensation, and filtering techniques.



Figure 24: Prana PAS-01

7.4.8 Air Quality Sensor - Prana PAS-01 Outdoor

Based on our initial selection, we are using the Prana PAS-01 Outdoor air quality sensor. Further testing and verification details follow:

Reasons for Selection	High-accuracy air quality sensing, multiple pollutant detection (PM2.5, PM10, CO ₂ , NO ₂ , O ₃), industrial-grade reliability, low maintenance, and compatibility with microcontrollers (ESP32, Raspberry Pi).
Key Specs (Datasheet Values)	• Measured Parameters: PM1.0, PM2.5, PM10, CO₂, NO₂, O₃, temperature, humidity • Communication: UART (RS485 Modbus), WiFi, LTE options • Accuracy: $\pm 10\%$ for PM, ± 50 ppm for CO₂, $\pm 5\%$ for humidity • Operating Temperature: -20°C to 50°C • Power Requirements: 5 V (USB) or external 12 V DC • Enclosure: IP65-rated (weatherproof) • Weight: 250 g

Testing Performed:

S. No	Test	Procedure
1	Baseline Calibration:	The sensor was placed in a controlled indoor environment, and initial readings were compared with reference instruments.
2	Outdoor Exposure Test:	The sensor was deployed outdoors for 48 h to validate long-term stability and detect environmental interference.
3	Response Time Check:	A controlled air sample with elevated PM2.5 levels was introduced to observe sensor response time.
4	Humidity Impact:	The sensor was tested under varying humidity levels to check measurement drift.

Testing Results:

Parameter	Value
Baseline Accuracy (PM2.5)	$\pm 8\%$ (compared to reference)
Response Time	10 s (PM), 30 s (gas)
Humidity Influence	$\pm 3\%$ deviation in high-humidity conditions
Outdoor Stability	Less than $\pm 5\%$ drift over 48 hours

Table 10: Prana PAS-01 Outdoor Testing Results

Calibration: The sensor was calibrated using co-located reference-grade air quality monitors. A multi-point correction factor was applied based on statistical regression analysis.

Integration Details: The PAS-01 sensor communicates with the ESP32 via UART (RS485). It is mounted in an open-air enclosure with a rain shield to prevent direct water exposure.

Code Snippet:

```

1 #include <SoftwareSerial.h>
2
3 SoftwareSerial pas01(16, 17); // RX, TX
4
5 void setup() {

```



```
6   Serial.begin(115200);
7   pas01.begin(9600);
8 }
9
10 void loop() {
11     if (pas01.available()) {
12         String data = pas01.readStringUntil('\n');
13         Serial.println("Sensor Data: " + data);
14     }
15
16     delay(2000);
17 }
```

Error Analysis: Potential sources of error include humidity-induced drift, sensor aging, and interference from nearby pollution sources. Error mitigation involves periodic recalibration, housing design adjustments, and real-time data filtering.

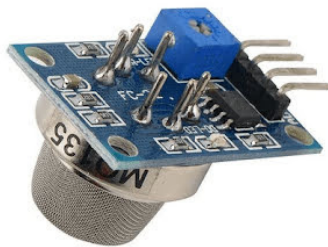


Figure 25: M135

7.4.9 Gas Sensor - M135

Based on our initial selection, we are using the M135 gas sensor. Further testing and verification details follow:

Reasons for Selection	High-sensitivity gas detection for CO, CO ₂ , NH ₃ , and other volatile organic compounds (VOCs). Suitable for air quality monitoring applications, good response time, easy integration with microcontrollers, and low power consumption.
Key Specs (Datasheet Values)	• Measured Gases: CO, CO₂, NH₃, VOCs • Operating Voltage: 5 V DC • Sensitivity: 10-1000 ppm (CO), 100-10000 ppm (NH₃) • Response Time: <30 s • Warm-up Time: <60 s • Communication: Analog output • Power Consumption: 150 mW • Operating Temperature: -10 °C to 50 °C • Weight: 15 g

Table 11: Specifications of the M135 Gas Sensor

Testing Performed:

S. No	Test	Procedure
1	Baseline Measurement:	The sensor was exposed to ambient air for 5 min to establish a reference baseline.
2	Gas Sensitivity Test:	The sensor was placed in a controlled chamber with known concentrations of CO and NH ₃ to measure response accuracy.
3	Response Time Check:	A sudden introduction of CO was used to observe the sensor's response time and recovery period.
4	Temperature Stability:	The sensor was exposed to varying temperatures from -10 °C to 50 °C to evaluate drift in gas readings.

Table 12: Testing Procedures for M135

Testing Results:

Parameter	Value
Baseline Drift	± 5% over 24 h period
Sensitivity Accuracy	± 10% (compared to reference values)
Response Time	<30 s
Temperature Influence	± 4% deviation over full range

Table 13: M135 Gas Sensor Testing Results

Calibration: The sensor is calibrated using known gas concentrations and a linear regression model. Periodic recalibration is recommended to maintain accuracy.

Integration Details: The M135 sensor is connected to an ESP32 microcontroller via an analog input pin. A voltage divider circuit is used to scale the sensor output appropriately.

Code Snippet:

```

1  #define SENSOR_PIN A0
2
3  void setup() {

```

```
4   Serial.begin(115200);  
5 }  
6  
7 void loop() {  
8     int sensorValue = analogRead(SENSOR_PIN);  
9     float voltage = sensorValue * (5.0 / 1023.0);  
10    Serial.println("Gas_Sensor_Voltage:" + String(voltage) + "V");  
11  
12    delay(1000);  
13 }
```

Error Analysis: Potential sources of error include sensor aging, humidity effects, and interference from other gases. Error mitigation includes frequent calibration, using a temperature-compensated model, and data smoothing techniques.



Figure 26: MQ131

7.4.10 Gas Sensor - MQ131

Based on our initial selection, we are using the MQ131 gas sensor module. Further testing and verification details follow:

Reasons for Selection	Need a reliable gas sensor to detect ozone (O_3) concentrations in the environment. Low power consumption, ease of use, and availability of breakout boards.
Key Specs (Datasheet Values)	• Sensor Type: Metal Oxide Semiconductor (MOS) sensor • Detection Target: Ozone (O_3) • Operating Voltage: 5 V • Current Draw: 150 mA (when heating) • Preheating Time: 24 hours for initial calibration • Resistance Range: 10 kΩ to 100 kΩ (depending on gas concentration) • Response Time: 30 seconds (for detection) • Weight: 20 g

Testing Performed:

S. No	Test	Procedure
1	Response Time Test:	We measured the time taken by the sensor to respond to ozone gas exposure by introducing known concentrations of ozone and observing the sensor output.
2	Sensor Calibration:	We calibrated the sensor in a controlled environment by exposing it to different ozone concentrations and recording the sensor resistance.
3	Noise Measurements:	We observed the noise level of the sensor in a stable environment with no gas exposure for 10 min.

Testing Results:

Parameter	Value
Sensor Response Time (to O_3)	30 seconds (typical)
Gas Concentration Range	10 ppb to 1 ppm
Heater Resistance (for 5V)	33 Ω
Gas Sensitivity (to O_3)	10 k Ω to 100 k Ω depending on concentration
Noise Level (no gas)	Minimal noise, stable resistance under no gas exposure
Calibration Sensitivity (O_3)	Linear with ozone concentration (within 10-1000 ppb range)

Table 14: MQ131 Testing Results

Error Analysis: Potential sources of error include sensor drift over time, temperature and humidity variations, and inconsistent gas exposure. To mitigate these errors, we are using a calibration procedure and environmental compensations (such as humidity sensors) to ensure accurate readings.

Integration Details: The MQ131 sensor is connected to the microcontroller (e.g., ESP32) via an analog input pin. It requires a heater circuit for its operation, and its output is measured through a voltage divider connected to an ADC (Analog-to-Digital Converter). The sensor is mounted on the central PCB.

Code Snippet:

```

1 int mq131Pin = A0; // Analog input pin for MQ131 sensor
2

```

```
3 void setup() {  
4     Serial.begin(115200);  
5 }  
6  
7 void loop() {  
8     int sensorValue = analogRead(mq131Pin);  
9     float ozoneConcentration = calculateOzoneConcentration(sensorValue);  
10    Serial.println("Ozone_Concentration:" + String(ozoneConcentration) + "ppb");  
11    delay(1000);  
12 }  
13  
14 float calculateOzoneConcentration(int sensorValue) {  
15     // Simple linear approximation for concentration  
16     return (sensorValue / 1023.0) * 1000.0; // Assume linear relationship  
17 }
```

Sensor Fusion: While the MQ131 sensor provides valuable data on ozone levels, we may fuse this information with other environmental data (e.g., temperature, humidity) to improve the overall accuracy of air quality monitoring. This could be done using a Kalman filter or other sensor fusion techniques.

Future Enhancements: Future work will involve implementing a more precise calibration model for the MQ131 sensor, along with the integration of a real-time clock (RTC) for time-stamped data collection.

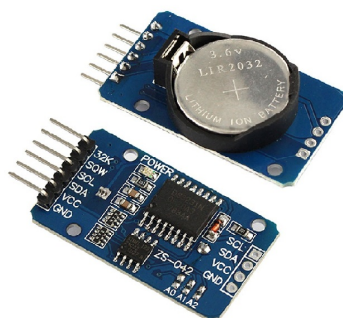


Figure 27: RTC

7.4.11 DS3231 - Real-Time Clock Module

Based on our initial selection, we are using the DS3231 RTC module. Further testing and verification details follow:

Reasons for Selection	High accuracy, I2C interface, Low power consumption, Easy integration with ESP32, Real-time clock functionality
Key Specs (Datasheet Values)	• Accuracy: ± 2 minute per year • Supply Voltage: 2.3 V to 5.5 V • Power Consumption: – Operating Mode: 1.0 mA – Standby Mode: 0.075 mA • Temperature Range: - 40 °C to 85 °C • Weight: 3.5 g • Interface: I2C

Testing Performed:

S. No	Component	Purpose
1	Accuracy Tests:	We compared the time kept by the DS3231 module against a reference clock. The test was conducted over a 24-hour period, checking for drift in the clock's accuracy.
2	Power Consumption Measurements:	The current draw of the DS3231 was measured using a multimeter in series with the power supply during both operating and standby modes.
3	Temperature Dependence Test:	The DS3231's performance was tested at various temperatures within its operating range. The temperature was varied from - 40 °C to 85 °C to observe any drift in accuracy.

Testing Results:

Parameter	Value
Accuracy (Time Drift)	1.5 minute per year (Average over 24 h)
Power Consumption (Operating Mode)	1.0 mA
Power Consumption (Standby Mode)	0.075 mA
Temperature Range Tested	- 40 °C to 85 °C

Integration Details: The DS3231 module is connected to the ESP32 via I2C. The SDA and SCL pins are connected to GPIO21 and GPIO22, respectively, of the ESP32. The module is mounted on the top plate of the CanSat.

Code Snippet:

```

1  #include <Wire.h>
2  #include <RTClib.h>
3
4  RTC_DS3231 rtc;
5
6  void setup() {
7      Serial.begin(9600);
8
9      // Check if the RTC is connected properly
10     if (!rtc.begin()) {
11         Serial.println("Couldn't find RTC");
12         while (1);
13     }

```

```

14
15 // Check if the RTC lost power
16 if (rtc.lostPower()) {
17     // Set the time to the compile time
18     rtc.adjust(DateTime(F(__DATE__), F(__TIME__)));
19 }
20 }
21
22 void loop() {
23     DateTime now = rtc.now();
24
25     Serial.print("Date/Time:␣");
26     Serial.print(now.year(), DEC);
27     Serial.print('/');
28     Serial.print(now.month(), DEC);
29     Serial.print('/');
30     Serial.print(now.day(), DEC);
31     Serial.print("␣");
32     Serial.print(now.hour(), DEC);
33     Serial.print(':');
34     Serial.print(now.minute(), DEC);
35     Serial.print(':');
36     Serial.print(now.second(), DEC);
37     Serial.println();
38
39     delay(1000); // Update every second
40 }

```

Error Analysis: Potential sources of error include temperature fluctuations, power supply variations, and I2C communication issues. The DS3231 is quite accurate, but external factors may still contribute to minor discrepancies.

Calibration Procedures: The DS3231 module is factory-calibrated for time accuracy. No specific calibration is required; however, the initial setup synchronizes the time based on the system's compile time.

7.5 Sensor Integration and Data Handling

- **Sensor Integration:** The integration of various sensors such as LPS22, AHT21B, and MPU9250 is accomplished through communication protocols like I2C and SPI, ensuring seamless data acquisition from each device.
- **Data Acquisition and Calibration:** Raw data from each sensor is calibrated to account for any measurement offsets or errors, ensuring that the sensors provide accurate and consistent readings over time.
- **Data Fusion:** Data from multiple sensors, such as the MPU9250 for tilt and rotation and the MS 4525DO for airspeed, is

fused using algorithms like Kalman filters to provide comprehensive insights into the environmental and motion parameters.

- **Data Logging and Storage:** All sensor data is logged in a structured format and stored using a memory storage system like an SD card, allowing for analysis and retrieval in future.
- **Data Processing and Analytics:** Collected data is processed to extract meaningful information, such as temperature and humidity patterns from the AHT21B or air quality readings from the Prana Pas 01, and used to generate reports and monitor conditions.
- **Redundancy and Failover:** Redundancy is built into critical sensors like BME680, BMP390, and MQ131 to ensure system reliability, using backup sensors in case of failure.
- **Data Visualization:** The final output is presented through user-friendly dashboards, aggregating data from all sensors and providing visualizations like graphs, tables, and maps to facilitate decision-making and track trends.

7.6 Sensor Data Flow

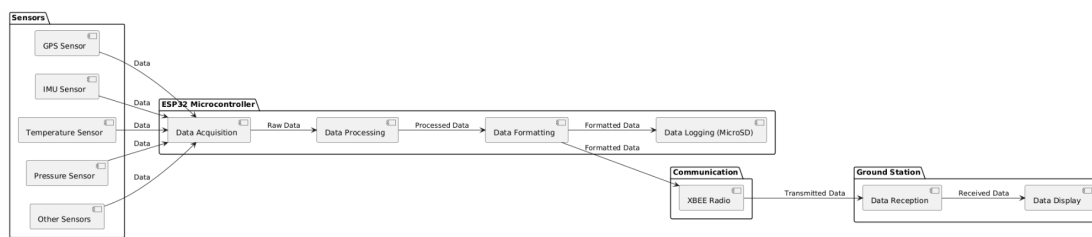


Figure 28: Sensor block diagram

Description: Data from various sensors (GPS, IMU, temperature, pressure, etc.) is collected by the ESP32 microcontroller. The ESP32 processes this data (unit conversions, filtering) and formats it for transmission. The processed data is then sent via the XBEE radio module to the ground station and stored in the microSD card.

7.7 Data Acquisition and Processing

- **ESP32 Data Reading:** The ESP32 reads data from each sensor using specific communication protocols (e.g., I2C, SPI, UART). The specific method depends on the sensor and its interface.
- **Data Processing:** The ESP32 performs necessary data processing steps, including:
 - **Unit Conversions:** Converting raw sensor readings to standard units (e.g., ADC values to volts, raw temperature to Celsius).
 - **Filtering:** Applying digital filters (e.g., moving average, Kalman filter) to reduce noise and smooth data.

- **Data Format for Ground Station:** The data is formatted into a structured packet for transmission. This could be a comma-separated value (CSV) string or a more complex format like JSON, depending on the requirements of the ground station software.

7.8 Power Management

Power to the sensors is managed by the ESP32. The ESP32 enables or disables power to individual sensors as needed to conserve power. For example, sensors that are not required during certain phases of the mission (e.g., GPS after landing) can be turned off.

Power Minimization Techniques:

- **Duty Cycling:** Sensors are turned on only when needed and turned off when not in use.
- **Low-Power Modes:** Using low-power modes of the ESP32 and sensors when possible.
- **Efficient Power Regulation:** Using efficient voltage regulators to minimize power loss.

7.9 Challenges and Mitigation Strategies

- **Noise Reduction:**
 - **Challenge:** Sensor readings can be noisy, affecting data accuracy.
 - **Mitigation:** Using digital filtering techniques, shielding sensitive components, and ensuring proper grounding.
- **Power Management:**
 - **Challenge:** Balancing power consumption with sensor performance.
 - **Mitigation:** Implementing duty cycling, using low-power modes, and optimizing power regulation.
- **Communication Problems:**
 - **Challenge:** Ensuring reliable communication between the CanSat and ground station, especially over long distances.
 - **Mitigation:** Using robust communication protocols, optimizing antenna placement, and testing communication links under various conditions.

8 Actuator Subsystem

8.1 Actuator Summary

8.1.1 Gyroscopic Wheel Motors



Figure 29: Gyroscope Motor

- Motor type: DC gear motor
- Operating voltage: 12V
- Weight: 19g
- Control method: Motor driver module with PWM Control
- Driver circuitry: Motor driver modules typically use H-bridge configurations to control the direction of the motor. [Motor Driver Ref.](#)
- Control algorithm:
 1. **Speed Control:** Pulse Width Modulation (PWM) is used to adjust the speed of the motor.
 2. **Direction:** The direction of the motor is controlled by changing the polarity of the voltage applied to the motor using the H-bridge circuit in the motor driver module.
 3. **Feedback Loop:** The feedback is taken from the IMU, motor driver is used to adjust the PWM signal to maintain the desired speed or position.

8.1.2 Parachute Deployment motor

- Motor type: [Servo Motor](#)



Figure 30: Servo Motor

- Operating voltage: 4.8 - 6.0 V
- Control interface: The ESP32 generates PWM signals to control the servo's position and speed.
- Torque specifications: $2kg - cm$
- Deployment mechanism:
 1. **Servo Horn:** A servo horn is attached to the shaft of the servo motor.
 2. **Linkage:** We have designed a latch mechanism in the parachute container, so when the servo rotates it turns open the parachute container lid, deploying the parachute.

8.1.3 System Level Gyroscopic Attitude Control

Operating Principles

- Three-axis stabilization approach using reaction wheels
- Wheel configuration: All the three wheel are placed mutually perpendicular to each other along the center axis of the cansat body.

8.2 Control System Logic

- Sensor input: Tilt and rotation data from the IMU.
- Control loop: We are using a PID Control for the 3 reaction wheels.
- Wheel speed management: After ejection, the CanSat delays gyro activation to avoid initial disturbances. Upon deploying the secondary descent mechanism, the bleed threshold is temporarily increased to 15° (from 3°) along with the bleed rate to counteract speed surges. Once stabilization is achieved, these values revert to normal, ensuring a controlled descent.

9 Payload/Gyroscope Subsystem

9.1 Payload Subsystem: Air Quality Sensor

We want to demonstrate that we can use the CanSat platform to perform detailed atmospheric analysis at a fraction of the traditional cost.

The slow descent rate of the cansat helps in collecting detailed and accurate data.

Our current work centers on testing and validating our air quality sensor setup, with a special emphasis on particulate matter measurements using the Prana Air PM2.5 sensor.

9.1.1 Dedicated Sensors and Particulate Matter Measurements

Our payload features a set of dedicated sensors designed to capture a variety of atmospheric parameters. At the heart of this setup is the Prana Air PM2.5 sensor, which we are testing extensively to ensure accurate detection of fine particulate matter. Alongside this, we are evaluating additional sensors to determine the best fit for our long-term deployment. The goal is to prove that a low-cost, reusable CanSat can perform sophisticated atmospheric measurements.

The other sensors we are using are the MQ135 and MQ131 (which measure NO_x and O₃)

9.1.2 Calibration and Validation

One of the key aspects of our project is ensuring the accuracy of our sensor data. To achieve this, we are not only calibrating our sensors in the lab but also using data from our college weather station as a field benchmark. By comparing our sensor outputs with the trusted readings from the weather station, we can fine-tune our measurements and correct for any biases. This dual calibration approach significantly enhances the reliability of our data.

9.1.3 Testing Methodology

Our testing process is designed to cover all bases:

- **Baseline Testing:** We start with comprehensive ground tests to ensure that each sensor is properly calibrated and working as expected. This step has been completed



Figure 31: Prana air PM 2.5

- **Altitude Trials:** Next, we will conduct altitude tests to observe how sensor readings change with altitude. This will help us understand the behavior of particulate matter and other atmospheric parameters at different heights.
- **Comparative Analysis:** We plan to test multiple sensors and compare their performance, ultimately selecting the one that offers the best accuracy and reliability for our needs.

9.1.4 Data Products and Expected Outcomes

Our sensor suite is expected to produce the following data:

- **Temperature vs. Altitude Graphs** – to show how temperature varies with height.
- **Humidity vs. Altitude Graphs** – to capture changes in moisture levels.
- **Pressure vs. Altitude Graphs** – including geolocation data.
- **PM2.5 vs. Altitude Graphs** – to map particulate matter distribution.
- **Gas Concentration vs. Altitude Graphs** – for monitoring trace gases.
- **Wind Speed/Direction Measurements** – to help determine the atmospheric boundary layer height.

9.1.5 Progress and Future Work

So far, our initial testing with the Prana Air PM2.5 sensor has been promising. The sensor is performing well under controlled conditions, and we are now gearing up for more extensive tests at various altitudes. In addition, by calibrating our sensors against data from our college weather station, we aim to ensure that our measurements are both accurate and reliable.

Looking ahead, our plans include:

- Conducting further alt tests to capture sensor behavior across different altitudes.
- Refining our calibration process using real-time data from the college weather station.
- Evaluating and comparing multiple sensor options to choose the best performer for our final design.

We are enthusiastic about the potential of this project and the valuable data it will provide.

9.2 Gyro Subsystem (Reaction Wheels)

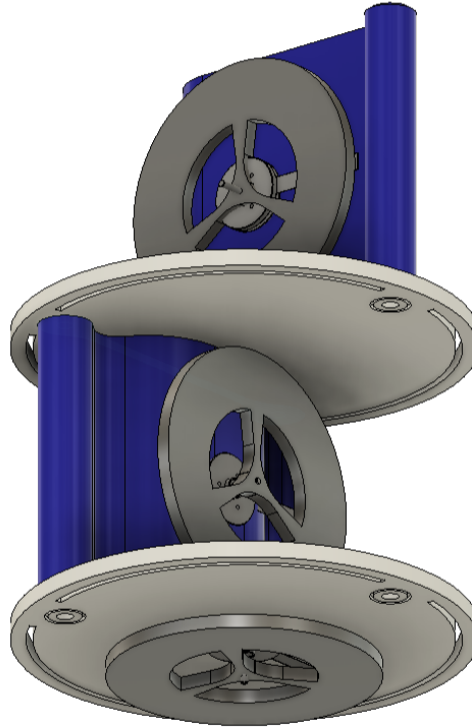


Figure 32: Gyroscope

CanSat Specifications:

- CanSat Height: 35 cm
- Distance from Center: 17.5 cm (approx.)
- Pitch/Yaw Moment of Inertia: $0.010\,162\text{ kg m}^2$
- Roll Moment of Inertia: $0.002\,328\text{ kg m}^2$

Motor Specifications:

- Maximum Speed: 1000 RPM
- Ring Diameter: 10 cm
- Ring Mass: 70 g (0.07 kg)

- Ring Moment of Inertia: $I = mr^2 = 0.000175 \text{ kg m}^2$ (Same for Roll, Pitch, and Yaw)
- Maximum Torque: $0.1 \text{ kg cm} \approx 0.01 \text{ N m}$

The motor used is the [GA12-N20 12V 1000 RPM DC Motor](#), chosen considering weight and power constraints. The motor weighs 19 g.

9.3 Control Algorithm and Desaturation Strategy

We must first address the issue of standing starts. When a motor starts from rest, the reaction time is very high since it takes a long time to spin up.

Options:

- Set a deadzone of $\pm 10\%$ RPM and spin in both directions.
- Default RPM = 55%, spin only in a single direction.

Considering we do not have a very high RPM motor, option (i) might be preferable. Although the deadzone is not ideal, a 10% RPM adjustment, if achieved slowly, should not adversely affect the balance of the CanSat.

For pitch and yaw, we will divide the tilt angle into components for the pitch and yaw gyroscopes. A simple PID algorithm will control the gyro. When the tilt is within 3° , the speed of the pitch and yaw motors can be bled very slowly. This approach is viable since the CanSat has some inertia and external forces such as airflow, parachute tugs, and potential fins that help maintain orientation.

Similarly, the roll gyro speed should also be bled when there is no rotational angular velocity, provided the roll and pitch gyros are also moving at low speeds. In cases of high saturation, the bleed threshold should be slightly increased to desaturate the motors faster.

Operation Strategy:

- Start the gyros a few seconds after the CanSat is ejected from the rocket.
- A few seconds after activating the secondary descent mechanism, increase the bleed threshold to 15° from the normal 3° and increase the bleed rate to dissipate the accumulated speed caused by the shock of deploying the parachute.
- Once the gyros are desaturated, revert the threshold and bleed rate to normal.

9.4 Gyro Wheel Weight and Design

A ring/wheel structure provides the highest moment of inertia per unit mass:

$$I = MR^2$$

To maximize I , the wheel structure should have the highest possible radius, with most of the mass concentrated at the outermost edge.

Why maximize I ? A higher moment of inertia increases the gyro system's saturation limit, making it proportional to I .

Wheel Weight Consideration: The wheels should use up as much of the weight budget as possible. If the wheels are too light, the gyro motor's maximum RPM should be prioritized over its torque.

9.5 Imbalance Scenario Analysis

Let us consider a scenario where a force equivalent to 30 g is applied at one end of the CanSat for 0.1 s.

$$F = 30 \times g = 0.03 \times 9.8 = 0.294 \text{ N}$$

$$\tau = F \times r = 0.294 \times 0.175 = 0.05145 \text{ N m}$$

Since the motor has a torque of 0.01 N m, the torque ratio for a 30 g force at the furthest end of the CanSat is:

$$\text{Torque Ratio} = \frac{0.05145}{0.01} = 5$$

Therefore, for 0.1 s of imbalance, the motor must run for 0.5 s to counteract the disturbance.

9.6 Angular Speed Calculation

Using the equation of motion:

$$I\alpha = \tau$$

$$\alpha = \frac{0.01}{0.000175} = 57.1429 \text{ rad/s}^2$$

$$\omega = \alpha \times t = 57.1429 \times 0.5 = 28.5 \text{ rad/s}$$

$$\text{Converting to RPM: } \omega \times \frac{60}{2\pi} = 272 \text{ RPM}$$

10 Descent Control Subsystem

10.1 Overview

This section explains how the descent control system works in our CanSat design. It covers the parachute system, how it is deployed, how it maintains stability, and how well it performs during descent.

10.2 Parachute System

The parachute is the main component that slows down the CanSat as it falls. It has a diameter of 120 cm and includes a 12 cm hole in the center, called a spill hole, which helps to keep it stable and prevent excessive swinging. The parachute is made from a carefully chosen material and designed to balance drag force and stability during descent.

10.3 Deployment Mechanism

The parachute must open at the right time to ensure a safe descent. To achieve this, we use a deployment system that stores the parachute in a small container and releases it at a specific altitude. The release mechanism uses both altitude sensors and a mechanical trigger to ensure the parachute opens properly and does not get tangled.

10.4 Stability and Performance Analysis

To evaluate how well the parachute system works, we measure how fast the CanSat falls, how stable it remains, and how much it drifts while descending. The descent speed is affected by the parachute's ability to create drag and slow down the CanSat. We also observe if the CanSat spins, swings, or moves unpredictably during the descent.

The expected terminal velocity (final constant falling speed) is calculated using the following equation:

$$mg = \frac{1}{2}\rho v_t^2 C_D A \quad (1)$$

where:

- $m = 1.0$ kg is the weight of the CanSat,
- $g = 9.81$ m/s² is the force of gravity,
- $\rho = 1.225$ kg/m³ is the air density,
- $C_D = 1.5$ is the estimated drag coefficient for a hemispherical parachute,
- $A = \pi(0.6)^2 \approx 1.13$ m² is the parachute's effective area.

Rearranging the equation to solve for terminal velocity:

$$v_t = \sqrt{\frac{2mg}{\rho C_D A}} \quad (2)$$

Substituting the values:

$$v_t \approx \sqrt{\frac{2 \times 1.0 \times 9.81}{1.225 \times 1.5 \times 1.13}} \approx \sqrt{\frac{19.62}{2.0779}} \approx \sqrt{9.44} \approx 3.07 \text{ m/s} \quad (3)$$

10.5 Enhancements and Future Improvements

During testing, the actual descent speed was higher than the calculated value, indicating that the parachute was not fully inflating or was not maintaining the ideal shape. To improve performance, we are planning the following modifications:

- Adjusting the way the parachute is attached to keep its shape consistent.
- Adding small stabilizing cords to reduce swinging.
- Using a swivel connector to prevent the CanSat from spinning too much.
- Testing a larger parachute to slow down the descent further.

11 Electrical Power Subsystem

11.1 Overview

Component	Purpose
Power	18650 Li-ion battery pack (3S) using Samsung INR18650-30Q cells (3.6 V, 3000 mAh each). Total energy: 32.4 Wh. A 3.3 V regulator powers most components; 5 V rail available if needed.
MCU	ESP32 serves as the primary controller for data acquisition and communication. Includes MicroSD shield for onboard storage and RTC module for mission timing.
Sensors	Includes Ublox Neo, BME680, LPS22HB, AHT21B, MPU9250, Prana Air Outdoor, MQ135, MQ131, ADS1115, INA260, and PX4. Powered by the 3.3 V rail. Power consumption calculations detailed below.
Actuators	DC motor (Maxon EC45) and servo for parachute deployment. Motor requires 12 V boost conversion.
Communications	XBee 3 and GSM modules handle data transmission at 3.3 V.
Audio & LED	100 dB buzzer provides audible alerts. LED indicators show system status via 3.3 V GPIOs.

Table 15: Overview of Power and Component Distribution

11.2 Voltage Conversion and Transient Suppression

We use an LM2596 buck regulator to convert from 12 V (supplied by a boost converter for the motor) down to 3.3 V for most electronics. In addition, three 2200 μF capacitors are connected in parallel with the battery to mitigate current spikes (e.g. during motor startup or GSM transmission).

11.3 Sensor Power Consumption Analysis

For each sensor, power is calculated using:

$$P = V \times I.$$

Below are detailed calculations for several representative sensors:

11.3.1 MPU-9250 (IMU)

- Operating Voltage: 2.5 V
- Current: 3.4 mA
- Calculation:

$$P = 2.5 \text{ V} \times 3.4 \text{ mA} = 8.5 \text{ mW}.$$

11.3.2 BMP390 (Pressure Sensor)

- Operating Voltage: 3.3 V
- In High Accuracy Mode: $I = 1.3 \mu\text{A}$
- Calculation:

$$P = 3.3 \text{ V} \times 1.3 \mu\text{A} \approx 4.29 \mu\text{W}.$$

11.3.3 BME680 (Multi-sensor: Pressure, Humidity, VOC)

- Operating Voltage: 3.3 V
- In combined mode: $I \approx 3.7 \mu\text{A}$
- Calculation:

$$P = 3.3 \text{ V} \times 3.7 \mu\text{A} \approx 12.21 \mu\text{W}.$$

11.3.4 AHT21B (Temperature & Humidity)

- Operating Voltage: 3.3 V
- Current: 0.98 mA
- Calculation:

$$P = 3.3 \text{ V} \times 0.98 \text{ mA} \approx 3.23 \text{ mW}.$$

11.3.5 GPS (NEO-M8N)

- Operating Voltage: 3.3 V
- Current: 23 mA
- Calculation:

$$P = 3.3 \text{ V} \times 23 \text{ mA} \approx 75.9 \text{ mW}.$$

11.3.6 Prana Air Outdoor (PM2.5 Sensor)

- Operating Voltage: 3.3 V
- Assumed Current: 15 mA
- Calculation:

$$P = 3.3 \text{ V} \times 15 \text{ mA} \approx 49.5 \text{ mW}.$$

11.3.7 MQ135 (Gas Sensor)

- Operating Voltage: 3.3 V
- Assumed Heater Current: 100 mA
- Calculation:

$$P = 3.3 \text{ V} \times 100 \text{ mA} = 330 \text{ mW}.$$

11.3.8 MQ131 (Ozone Sensor)

- Operating Voltage: 3.3 V
- Assumed Current: 80 mA
- Calculation:

$$P = 3.3 \text{ V} \times 80 \text{ mA} = 264 \text{ mW}.$$

11.3.9 ADS1115 (Voltage Sensor)

- Operating Voltage: 3.3 V
- Assumed Current: 0.2 mA
- Calculation:

$$P = 3.3 \text{ V} \times 0.2 \text{ mA} \approx 0.66 \text{ mW} \quad (\text{rounded to } 1 \text{ mW}).$$

11.3.10 INA260 (Current/Voltage Sensor)

- Operating Voltage: 3.3 V
- Assumed Current: 1 mA
- Calculation:

$$P = 3.3 \text{ V} \times 1 \text{ mA} = 3.3 \text{ mW}.$$

11.3.11 LPS22HB (Pressure Sensor)

- Operating Voltage: 3.3 V
- Assumed Current: 0.5 mA
- Calculation:

$$P = 3.3 \text{ V} \times 0.5 \text{ mA} = 1.65 \text{ mW}.$$

11.3.12 Dedicated Humidity Sensor

- Operating Voltage: 3.3 V
- Assumed Current: 1 mA
- Calculation:

$$P = 3.3 \text{ V} \times 1 \text{ mA} = 3.3 \text{ mW}.$$

11.4 Battery Details

Battery Configuration:

- 3 cells in series; each cell: 3.6 V, 3000 mAh.
- Total Nominal Voltage:

$$V_{\text{total}} = 3 \times 3.6 \text{ V} = 10.8 \text{ V}.$$

- Total Energy:

$$E_{\text{battery}} = 3 \times (3.6 \text{ V} \times 3 \text{ Ah}) = 32.4 \text{ Wh.}$$

11.5 Total Power Consumption of All Components

The following table aggregates the power draws of each subsystem based on the calculations above. (Note: These values are approximate and assume simultaneous operation.)

Subsystem	Components	Total Power Draw (W)
Sensors	MPU-9250, BMP390, BME680, AHT21B, GPS, Prana Air Outdoor, MQ135, MQ131, ADS1115, INA260, LPS22HB, Dedicated Humidity Sensor, PX4	$\approx 0.74 \text{ W}$
Microcontrollers	ESP32, MicroSD Shield, RTC (DS3231)	$\approx 0.28 \text{ W}$
Communications	XBee 3, GSM, Antenna	$\approx 0.055 \text{ W}$
Motor & Driver	Maxon EC45 Motor (15 W) + Driver (0.024 W)	$\approx 15.024 \text{ W}$
Total (Full System)	Sum of Above	$\approx 16.1 \text{ W}$

Table 16: Aggregate Power Consumption of All Subsystems

11.6 Battery Runtime Analysis

Using the 32.4 Wh battery, we analyze the mission in three phases:

11.6.1 Phase 1: Standby Mode (Sensors + ESP32 Only)

- Assumed Power Draw: 1 W (low-power operation)
- Duration: 2 hours
- Energy Consumption:

$$E_{\text{standby}} = 1 \text{ W} \times 2 \text{ h} = 2 \text{ Wh.}$$

11.6.2 Phase 2: Flight Mode (Full System with Motors Active)

- Assumed Power Draw: 15.38 W (dominated by motor and active components)
- Duration: 0.5 hours (30 minutes)
- Energy Consumption:

$$E_{\text{flight}} = 15.38 \text{ W} \times 0.5 \text{ h} \approx 7.69 \text{ Wh.}$$

11.6.3 Phase 3: Recovery Mode (Low-Power with Active Buzzer)

- Assumed Power Draw: 1.1 W (includes slight additional draw for the buzzer)
- Duration: 0.5 hours (30 minutes)
- Energy Consumption:

$$E_{\text{recovery}} = 1.1 \text{ W} \times 0.5 \text{ h} = 0.55 \text{ Wh.}$$

11.6.4 Total Mission Energy Consumption

$$E_{\text{total}} = E_{\text{standby}} + E_{\text{flight}} + E_{\text{recovery}} = 2 + 7.69 + 0.55 \approx 10.24 \text{ Wh.}$$

$$\text{Battery Reserve} = 32.4 \text{ Wh} - 10.24 \text{ Wh} \approx 22.16 \text{ Wh} \quad (\approx 68\% \text{ margin})$$

11.7 Conclusion

The detailed calculations show that:

- 2 hours of standby (low-power mode) consumes 2 Wh.
- 30 minutes of flight (full system with motors active) consumes about 7.69 Wh.
- 30 minutes of recovery (with an active buzzer) consumes roughly 0.55 Wh.

The total energy consumption for the mission is approximately 10.24 Wh, which is significantly lower than the 32.4 Wh provided by the battery pack. This results in a reserve margin of about 68%, ensuring robust operation even accounting for inefficiencies and real-world variations.

12 Flight Software & Ground Control

Software Subsystem

12.1 Overall Overview

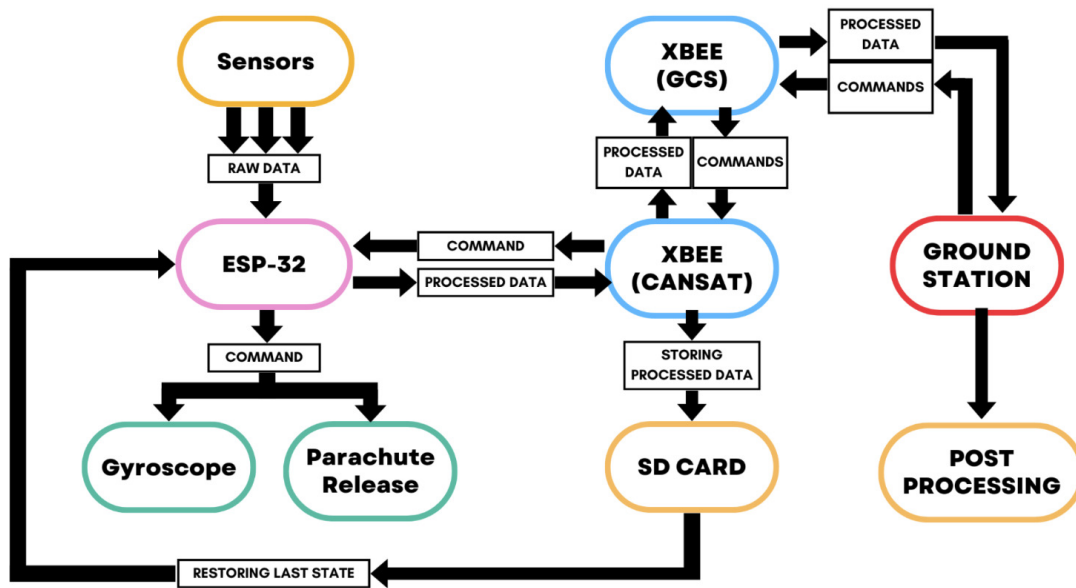


Figure 33: Overall Data Flow Chart

12.2 Flight Software

The flight software is responsible for managing and controlling the various subsystems of the CanSat. It is integrated with the communication subsystem, allowing it to transmit and receive data from the ground station.

12.3 FSW Objectives

- Upon power on, the flight systems will be initialized.
- Before the launch, the CanSat time will be set within 1 second of UTC (or the mentioned time) upon receiving the Set Time command from the Ground Control Station (GCS).
- Prior to the launch, upon receiving the altitude calibration command, the altitude will be calibrated to 0 m.

- Handle processor resets without significantly affecting flight operations.
- Upon landing, the CanSat shall stop data transmission and activate an audio beacon.
- In the simulation mode:
 - Receive barometric data from GCS and produce output for altitude according to the given barometric value.

12.4 Programming and Development Environment

- **Programming Language:** C/C++
- **Development Environment:** Visual Studio Code, Arduino IDE

12.5 FSW Overview

12.5.1 Flight Mode

- **Sensor:** ESP-WROOM-32
- **Communication:** XBEE and GSM 808
- **Storage:** SD Card (On-board storage)
- **Data Sampling Rate:** 10 Hz (Raw data)
- **Processed Data Rate:** 1 Hz
- **Commands:** Activation Command, Pressure Data (1 Hz)
- **Last Save Point:** In case of processor reset

12.6 Simulation Mode

- Similar communication and data flow as Flight Mode
- Barometric data received from GCS and altitude output produced

12.7 FSW State Diagram

- State 0: CANSAT on standby (turned on)
- State 1: CANSAT Test Mode

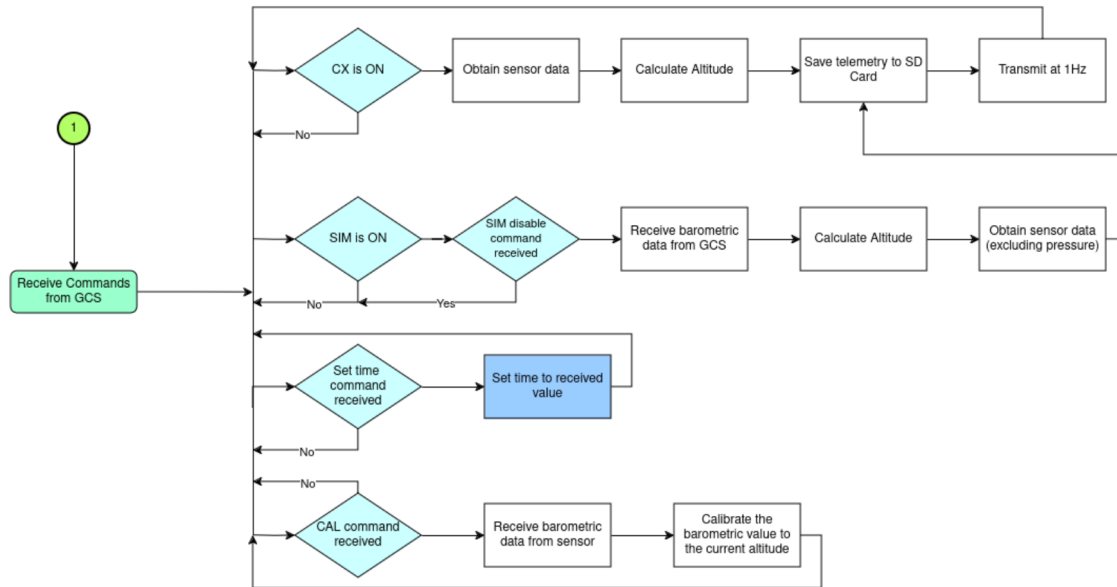


Figure 34: Simulation State Diagram Flowchart

- State 2: CANSAT Launchpad mode
- State 3: Ascending state
- State 4: Deployed State
- State 5: Parachute 1 Deployed State
- State 6: Parachute 2 Deployed State
- State 7: Landing Preparation State
- State 8: Impact

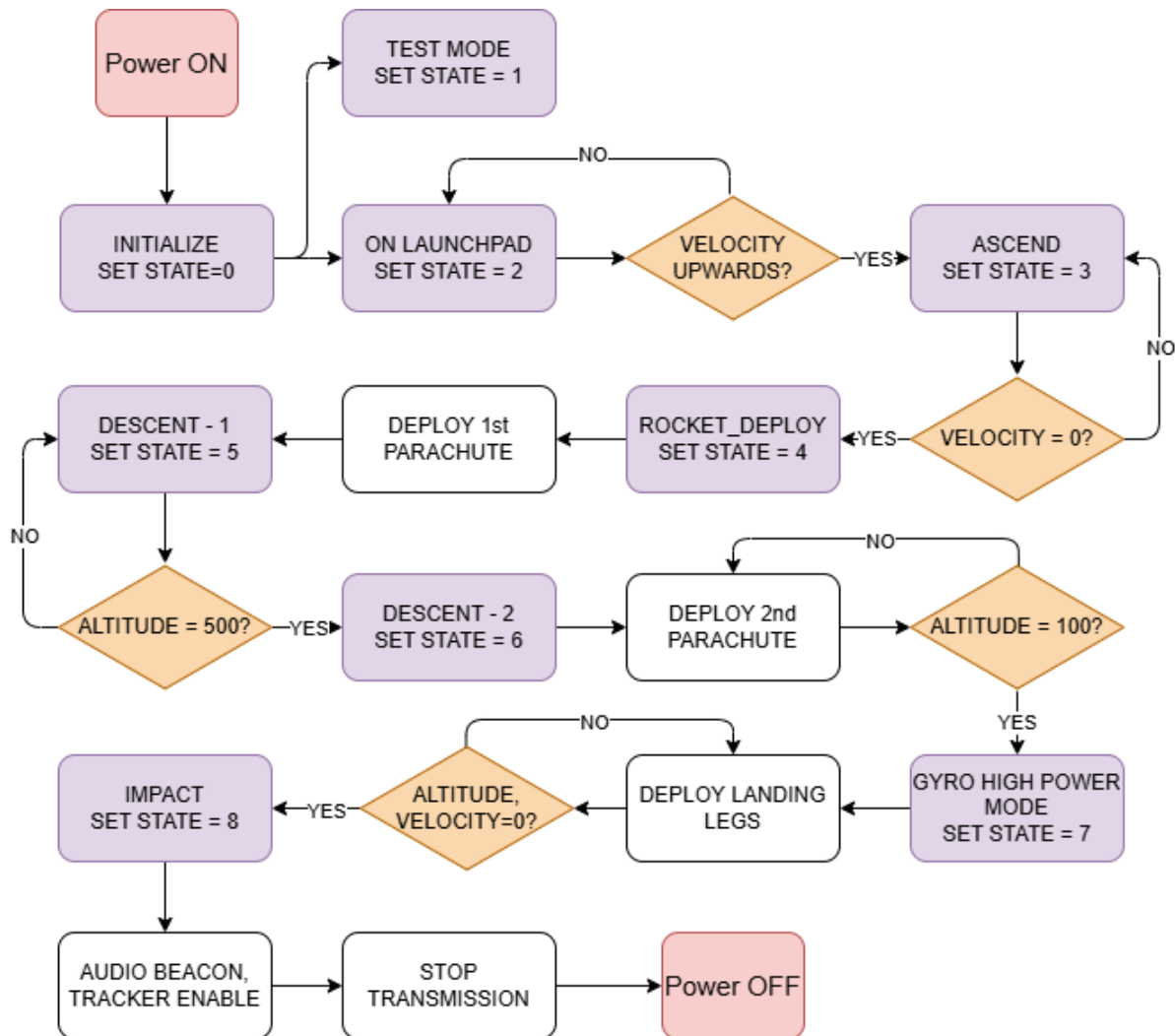


Figure 35: State Diagram Flowchart

12.8 Data Recovery Plan

- Flight Software State saved on SD Card for recovery in case of software reset.
- Critical data stored in CSV format on on-board storage:
 - Mission time, altitude calibration data, number of packets transmitted
 - Most recent command received, flight mode, parachute and heat shield status
 - Telemetry status
- Data recovery ensures seamless resumption of flight operations post-reset.

12.9 FSW Testing

TEST	TEST DESCRIPTION	REQUIREMENT TO PASS	STATUS
Flight Mode Test	Put the CanSat into Flight mode for 2 minutes, then turn CanSat On and Off. Confirm the FSW has returned to correct packet count, flight state, mission time and zero calibration value.	Maintains packet count, flight state, mission time and zero calibration post processor reset.	PASS
Mission Time Test	The ST command is sent to CanSat	The time sent by CanSat telemetry should be within 1s off from the original UTC time	PASS
Flight Data Inspection Test	Power the CanSat and GCS. Enter the Flight mode, send CX, ON command to payload to begin telemetry. Confirm the telemetry is sent according to Mission guide specifications.	Calibrates telemetry to zero post CX, ON. Delivers packets with all telemetry fields as specified by mission guide.	PASS
Simulation Mode Data Test	Power the CanSat and GCS. Enter the simulation mode using SIM ENABLE and ACTIVATE commands. Send simulation pressure commands and confirm CanSat reads and acts according to given pressure values.	CanSat reads simulation command and enters simulation mode and returns telemetry based on input pressure values.	PASS

12.10 Ground Control System (GCS)

The GCS consists of two software modules that operate simultaneously and independently.

12.10.1 Communication

XBEE S2C with inbuilt antenna had lesser range as compared to using XBEE Pro S2C. We will use a 3AN High Gain 12 dBi omni-directional WiFi Router Antenna for better gain.



12.10.2 Linker Interface

This module reads and processes serial data from the ground-based communication system.

Key Functions:

- Parses and formats incoming data.
- Stores data in a CSV file for later analysis.
- Exposes data via an API for real-time dashboard visualization.

Implementation Details:

- **Language:** Python
- **Status:** Complete
 - Simulation with random values: Complete
 - Testing with communication system: In progress
- **Chances of modifications:** Low

12.10.3 Dashboard

The dashboard provides an intuitive and real-time view of critical CanSat data.

Key Features:

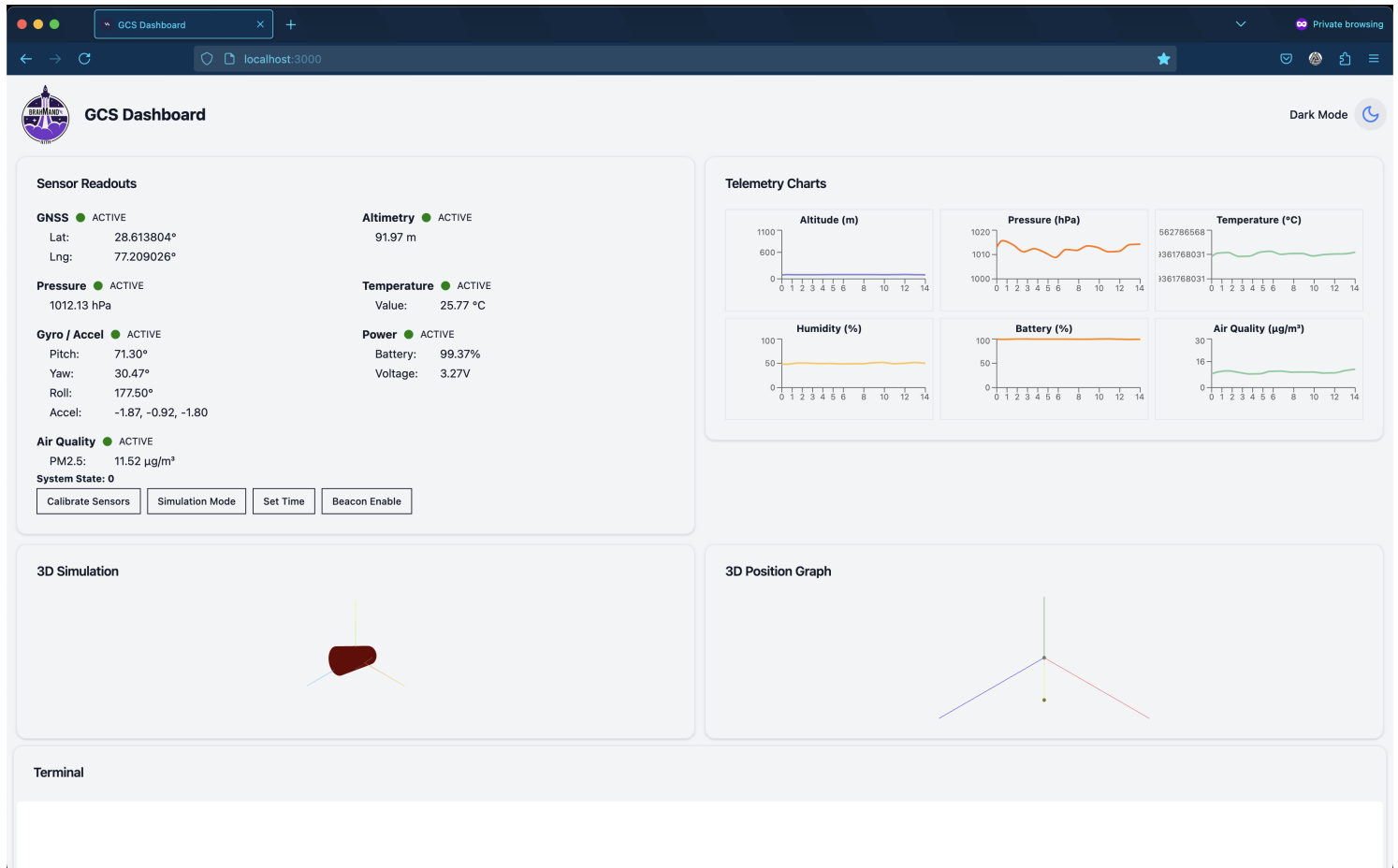


Figure 36: Dashboard Interface for CanSat Ground Control

- Displays essential telemetry data in an easy-to-understand format.
- Shows the status of all sensors.
- Includes a 3D graphical view to visualize the CanSat's orientation and location.
- Provides graph views of core sensor values for real-time data analysis.
- Displays the current location and operational status of the CanSat.
- Features an integrated terminal for sending and receiving signals (in progress).
- Retrieves data dynamically from the API, enabling distributed operation across multiple machines.

Implementation Details:

- **Language:** Next.js
- **Modules:** Three.js, Three Fiber
- **Status:** Complete
 - Simulation with random values: Complete
 - Testing with communication system: Ongoing
 - Fine-tuning of 3D model and UI/UX: Ongoing
- **Chances of modifications:** Medium

12.11 GCS Subsystem

TEST	TEST DESCRIPTION	REQUIREMENT TO PASS	STATUS
GCS Interface and Data Inspection, Format Test	Turn on the CanSat and send CX, ON command. Monitor the telemetry data received.	The GCS Interface displays all the telemetry data received (graphs and labels) and is presented in engineering units and saved into a .csv file.	PASS
Audio Beacon and Telemetry Test	Set CanSat to Simulation mode, and send pressure values corresponding to launch pad altitude.	The CanSat telemetry should stop and the audio beacon should start.	ONGOING
Data Packet Loss Test	Have an LED indication whenever the CanSat module sends data, and an LED indication when the ground station receives a packet.	Manually verify if the packet loss counted by the ground station is the same as the difference in number of data rows in the telemetry on board and received. This should also be consistent with the difference in LED blinks.	ONGOING

12.12 Comprehensive Failure Modes and Analysis

12.12.1 Communication Failures

- **Signal Interference:** High-altitude environments may experience electromagnetic interference
 - *Mitigation:* Implement error correction codes and packet verification
 - *Testing:* Conduct tests in noisy RF environments to verify robustness
- **Antenna Misalignment/Damage:** Physical orientation changes during flight can affect signal strength
 - *Mitigation:* Redundant antennas or omnidirectional antenna design
 - *Testing:* Test communication at various orientations and distances

12.12.2 Power System Failures

- **Battery Depletion:** Cold temperatures at altitude can reduce battery performance
 - *Mitigation:* Thermal insulation for batteries, power management algorithms
 - *Testing:* Cold-environment battery discharge testing
- **Power Spikes/Brown-outs:** Launch vibration can cause connection issues
 - *Mitigation:* Power conditioning circuits, capacitor buffers
 - *Testing:* Vibration testing with power monitoring

12.12.3 Sensor Failures

- **Barometric Sensor Drift:** Pressure sensors can provide false readings under rapid pressure changes
 - *Mitigation:* Implement Kalman filtering, sensor fusion with IMU data
 - *Testing:* Pressure chamber tests with rapid pressure changes
- **IMU Saturation:** Gyroscopes/accelerometers can saturate during high-g events
 - *Mitigation:* Use appropriate sensor ranges, implement detection algorithms
 - *Testing:* Centrifuge testing for high-g tolerance

12.12.4 Software Failures

- **Deadlock/Race Conditions:** Multi-threaded operations can cause system freezes
 - *Mitigation:* Watchdog timers, critical section protection
 - *Testing:* Stress testing with randomized command sequences
- **Memory Corruption:** Cosmic rays or power issues can corrupt memory
 - *Mitigation:* Error-correcting memory, periodic memory verification
 - *Testing:* Memory corruption simulation tests

12.12.5 Environmental Failures

- **Thermal Management:** Components may fail outside operating temperature ranges
 - *Mitigation:* Thermal insulation, heaters for critical components
 - *Testing:* Thermal chamber testing across expected altitude temperatures
- **Moisture/Condensation:** Rapid altitude changes can cause condensation
 - *Mitigation:* Conformal coating, moisture barriers
 - *Testing:* Thermal cycling with humidity control

12.12.6 Mechanical Failures

- **Vibration-Induced Connection Failures:** Launch vibrations can loosen connections
 - *Mitigation:* Thread-locking compounds, vibration-resistant connectors
 - *Testing:* Shake table tests at expected launch frequencies
- **Parachute Deployment Failures:** Critical for safe landing
 - *Mitigation:* Redundant deployment mechanisms, multiple trigger conditions
 - *Testing:* Deployment testing in various orientations and conditions

12.13 Extended Testing Matrix

TEST CATEGORY	TEST DESCRIPTION	EXPECTED BEHAVIOR	REQUIRED EQUIPMENT	PASS CRITERIA
Communication Resilience	Introduce signal attenuation between CanSat and GCS in 5dB increments	System should maintain data integrity with increased error correction	Variable RF attenuator, Spectrum analyzer	<25% packet loss at -90dBm
Range Testing	Test maximum communication range in open field conditions	Determine practical operation range and signal degradation pattern	GPS for distance measurement, Tripod mount	Reliable connection at competition-required distances
Failover Testing	Simulate primary processor failure during operation	Secondary processor should take over with minimal data loss	Hardware fault injection tools	<5 second recovery time, <10% data loss
Power Disruption	Simulate momentary power loss/brownout conditions	System should recover and resume operation from last saved state	Programmable power supply	Resume correct state with data integrity
Out-of-bounds Sensor Data	Feed sensors with values outside expected ranges	Software should detect anomalous data and use fallback algorithms	Sensor simulation jig	No crashes, appropriate error flags set
Memory Corruption	Corrupt random memory locations during operation	Error detection should identify issues and recover when possible	JTAG debugger	System stability maintained, errors logged
Environmental Chamber	Subject assembled CanSat to temperature cycling from -10°C to 50°C	All systems should function within specification across temperature range	Environmental chamber	All sub-systems operational across temperature range

TEST CATEGORY	TEST DESCRIPTION	EXPECTED BEHAVIOR	REQUIRED EQUIPMENT	PASS CRITERIA
Vibration Testing	Subject CanSat to launch-like vibration profile	No mechanical or electrical failures	Vibration table	No loose connections, all systems operational post-test
Recovery System	Simulate multiple deployment failure scenarios	System should attempt alternative deployment methods	Deployment mechanism test rig	Successful deployment under at least 2 failure conditions
Low Battery Behavior	Test system behavior as battery depletes	Graceful degradation with critical functions prioritized	Programmable load, Battery monitor	Critical systems remain functional below 30% power

Table 17: Extended Testing Matrix for CanSat Failure Modes

12.14 Software Improvement Recommendations

- State Machine Documentation:** Create a detailed state transition table showing all possible transitions and edge cases.
- Health Monitoring System:** Implement a dedicated health monitoring subsystem that continuously checks all critical components and reports status.
- Graceful Degradation Logic:** Program the system to prioritize mission-critical functions when resources are limited (low power, damaged sensors, etc.).
- Automated Testing Framework:** Develop automated tests that can verify system behavior across all states and transitions.
- Telemetry Compression:** Implement data compression to maximize bandwidth usage during critical flight phases.

12.15 FMEA (Failure Mode and Effects Analysis)

COMPONENT	FAILURE MODE	EFFECT	SEVERITY (1-10)	PROBABILITY (1-10)	DETECTION DIFFICULTY (1-10)
Primary Processor	Complete Failure	Loss of control	10	3	2
Primary Processor	Partial or Intermittent Failure	Unpredictable behavior	8	4	6
Communication Module	Signal Loss	Loss of telemetry	7	5	4
Communication Module	Data Corruption	Incorrect commands	8	4	7
Power System	Complete Failure	Mission failure	10	2	2
Power System	Partial Power	Limited functionality	6	4	5
Barometric Sensor	Drift	Incorrect altitude	7	6	8
Barometric Sensor	Complete Failure	No altitude data	8	3	3
IMU	Drift	Incorrect orientation	5	7	8
IMU	Complete Failure	No orientation data	6	3	3
SD Storage	Write Failure	Data loss	7	4	6
Primary Parachute	Deployment Failure	High-velocity impact	10	3	4
Secondary Parachute	Deployment Failure	High-velocity impact	10	2	4

Table 18: FMEA for CanSat Critical Components

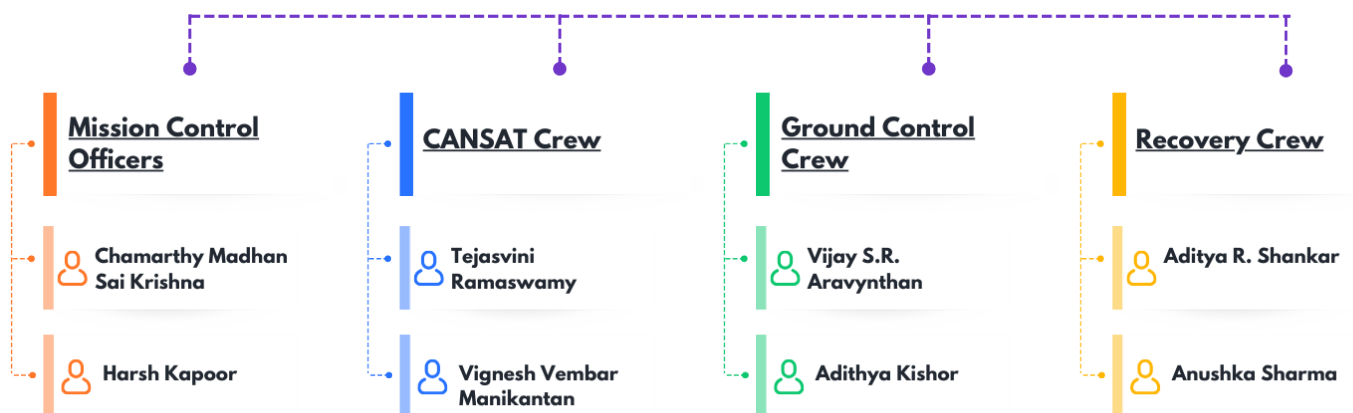
13 Mission Operation & Analysis

13.1 Overview of Mission Sequence of Events

On launch day, every team member will actively contribute to the success of the mission. A Mission Operations Manual will be created and it will serve as a comprehensive guide, outlining the key events and protocols that all team members must follow. Collaboration is essential throughout the mission, ensuring seamless coordination of tasks whenever possible. The following table summarizes the roles and responsibilities of the team members and the teams they will be divided in.

Role	Team Member Name	Responsibilities
Mission Control Officers	Harsh, Madhan	Supervision of all Departments, POC on the field with flight coordinator
CanSat Crew	Tejasvini, Vignesh	Preparation, Assembly and Integration of CanSat to the rocket
Ground Control Crew	Vijay, Adithya	Setup and operation of Ground Control Station and Monitoring CanSat telemetry
Recovery Crew	Aditya, Anushka	Tracking and retrieval of CanSat after Launch and interacting with the field judges

The following chart also gives the team division for the launch day activities -



13.2 Launch Day Sequence of Events

The following table explains the sequence of events, as they occur on the launch day and the teams which perform them -

Event	Tasks	Team Involved
Arrival	Arrival at the launch site with CanSat and Ground Station Hardware	ALL
Pre-Launch Events	Assemble CanSat and Mechanical System Integration	Cansat Crew
	Ground Station Setup and Antenna Calibration	Ground Control Crew
	Calibrate and Test Electrical Systems	Cansat Crew
	Turn CanSat on	Cansat Crew
	CanSat Integration into Rocket	Cansat Crew
Launch Events	Move to Launch Control Table	ALL
	Initiate CanSat Launch	Mission Control Officer
Descent	Monitor the Data Reception from Telemetry and Validate Flight State Changes	Ground Control Crew
Recovery Phase	Locate and Recover the CanSat using GPS and Audio-Visual Trackers	Recovery Crew
Submission Phase	Submit the CanSat, SD Cards, and Flight Data to the Field Judge	Ground Control Crew, Recovery Crew
Data Analysis	Analyze the Flight and Telemetry Data	Ground Control Crew
	Post Flight Review (PFR)	ALL
	Return to Check-in	Recovery Crew

13.3 Recovery Strategies

Recovery Strategy	Descriptions
Bright Coloring	Using Neon Orange and Green colored outer covering for the body.
	Using Parachutes of Bright Red color.
	Using bright filaments for 3-D printing.
Visual Beacon	Placing red LEDs which flash upon landing.
	Using Mylar streamer for active tracking from ground.
Audio Beacon	Using a beeping buzzer which activates upon landing.
GPS Position	Using Real-time GPS location from the GPS module of CanSat.
Team Contact	Placing the following card on the various CanSat modules so that in the unlikely event of CanSat being lost and found by someone else, it can be safely returned. It will have the information in English, Hindi and the regional Language of the launch site (here for placeholder it is Telugu)

Contact |సంపర్క|సంప్రదించండి

చిరునామా- ఇంటర్నేషనల్ ఇన్స్టిట్యూట్ ఆఫ్ ఇన్ఫర్మేషన్ టెక్నాలజీ,
 ప్రొఫెసర్ సి.ఆర్. రావు రోడ్,
 గచ్చిబౌలి, హైదరాబాద్ - 500032,
 తెలంగాణ, భారతదేశం

पता - अंतर्राष्ट्रीय सूचना प्रौद्योगिकी संस्थान,
 प्रो. सी. आर. राव रोड,
 गाचीबोवली, हैदराबाद - 500032,
 तेलंगाना, भारत

Address - International Institute of Information Technology,
 Prof. C. R. Rao Road,
 Gachibowli, Hyderabad - 500032,
 Telangana, India

Phone - 9305752443,
 6300696920



TEAM BRAHMAND
 IN-SPACE CANSAT

13.4 CANSAT Integration and Testing

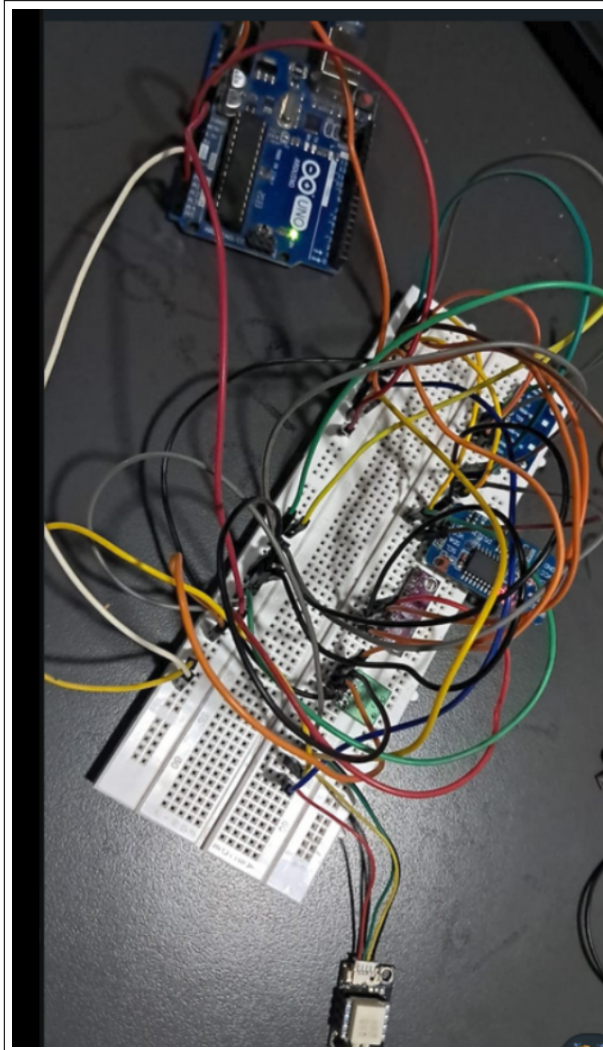
Testing Stage	Description
Subsystem Level Test Plan	Test individual components/subsystems (e.g., Sensors, CDH, EPS, GCS, etc.) to identify and troubleshoot before integrating into a larger system.
Integrated Level Test Plan	Testing for Integrated Systems (e.g., Descent Control, Electro-mechanical systems, Communications) to identify and troubleshoot any issues.
Environment Test Plan	With a fully integrated CanSat, we test the entire assembly according to environmental testing standards set by the mission guide.
Simulation Test Plan	The CanSat is commanded in Simulation mode, and the simulated launch data is sent to the Ground Station to demonstrate FSW, CDH.

The following sub-sections summarize the testing plans at different stages -

13.4.1 Subsystem Level Testing Plan

Subsystem	Testing Plan
Sensors	• Test and calibrate the sensor. • Test individually and verify sensor accuracy.
CDH	• Test communication between sensors and micro-controller. • Verify data logging and mid-flight recovery. • Ensure proper data formatting and transmission protocols.
EPS	• Test for 2-hour battery endurance. • Verify reverse polarity protection and voltage regulation for all components. • Verify reverse polarity protection and voltages for all sensors. • Test power consumption of individual subsystems.
Communications	• Test long-distance communication. • Test two-way communication between XBEEs.
FSW	• Test transitions between simulation and flight states in code. • Test flight protocol for start-up and reset. • Verify time-set code with GPS sensor.
Mechanical	• Test CanSat's structural integrity. • Test mechanisms functionality.
Descent Control	• Verify parachute mechanisms' functionality via drop tests. • Test descent rate control and stability. • Ensure proper functioning of secondary descent mechanism.

13.4.2 Sensor Integrated Testing



The following output represents the integrated testing of various sensors:

Integrated code testing:

PX4 -> Raw Pressure: 8111

PX4 -> Airspeed (m/s): 3.64

MPU9250 -> Sensor ID: 0

MPU9250 -> Accel X: 0.00

MPU9250 -> Accel Y: 0.00

MPU9250 -> Accel Z: 0.00

MPU9250 -> Gyro X: 0.00

MPU9250 -> Gyro Y: 0.00

```
MPU9250 -> Gyro Z: 0.00
MPU9250 -> Magnetometer error, reinitializing...
MPU9250 -> Mag X: 0.00
MPU9250 -> Mag Y: 0.00
MPU9250 -> Mag Z: 0.00
MPU9250 -> Horizontal Direction: 0.00
ADS1015 -> AIN0: 296 (0.89 V)
ADS1015 -> AIN1: 295 (0.88 V)
ADS1015 -> AIN2: 295 (0.88 V)
ADS1015 -> AIN3: 295 (0.88 V)
-----
AHT20 -> Temp: 29.51 °C, Humidity: 35.04 %
DS3231 -> Date/Time: 2024-2-19 0:2:0
PX4 -> Raw Pressure: 8106
PX4 -> Airspeed (m/s): 3.64
MPU9250 -> Sensor ID: 0
MPU9250 -> Accel X: 0.00
MPU9250 -> Accel Y: 0.00
MPU9250 -> Accel Z: 0.00
MPU9250 -> Gyro X: 0.00
MPU9250 -> Gyro Y: 0.00
MPU9250 -> Gyro Z: 0.00
MPU9250 -> Magnetometer error, reinitializing...
MPU9250 -> Mag X: 0.00
MPU9250 -> Mag Y: 0.00
MPU9250 -> Mag Z: 0.00
MPU9250 -> Horizontal Direction: 0.00
ADS1015 -> AIN0: 296 (0.89 V)
ADS1015 -> AIN1: 295 (0.88 V)
ADS1015 -> AIN2: 295 (0.88 V)
ADS1015 -> AIN3: 295 (0.88 V)
```

This output showcases the real-time data from the PX4, MPU9250, ADS1015, AHT20, and DS3231 sensors. It includes pressure readings, airspeed, accelerometer data, humidity, temperature, and the current date and time from the DS3231 RTC.

13.4.3 Integrated Level Functional Test Plan

Test Category	Test Plan
Descent Testing	• Ensure parachute descent speeds are accurate to 1 m/s of desired speeds. • The secondary descent control mechanism starts operating at the desired height (500m). • Test the deployment of the primary and secondary parachutes to ensure they function correctly.
Communications	• Test long-distance radio communication between XBEEs. • Test two-way communication and optimal antenna position. • Test Ground Control Station's GUI display and ensure real-time telemetry data is displayed correctly.
Mechanisms	• Test simulation mode software with given competition datasets. • Ensure simulation mode commands are transmitted through two-way communication. • Ensure simulation mode data is received by the ground station and processed correctly.
Deployment	• Test the parachutes to ensure they won't become tangled on release. • Test tolerances between parachutes, CanSat, and rocket to ensure smooth deployment. • Test the tensile strength of parachute cords via environmental drop tests.
Simulation	• Test simulation mode software with given competition datasets. • Ensure simulation mode commands are transmitted through two-way communication. • Ensure simulation mode data is received by the ground station and displayed correctly.

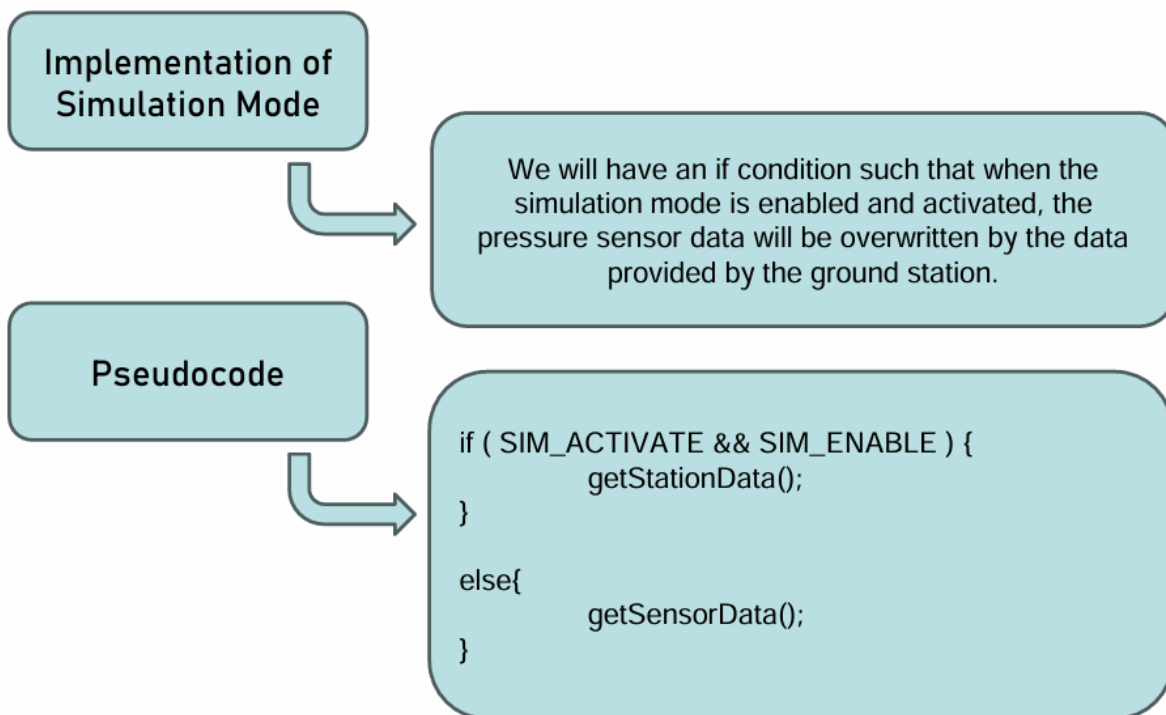
13.4.4 Environmental Test Plan

Test Category	Test Plan
Drop Test	• Ensure Cansat can survive 30Gs of shock • Attach the parachute and drop it to shock the Cansat leaving the rocket body
Thermal Test	• Ensure that the CanSat operates successfully in a 60°C environment. • Build a thermal chamber and place the functioning CanSat inside at 60°C for two hours.
Vibration Test	• Verify the mounting integrity of all components, mounting connections, structural integrity, and battery connections. • Attach the CanSat to an orbital sander and cycle it for five intervals of five seconds.
Fit Check	• Ensure the CanSat fits in the rocket and deploys without issues. • Construct a rocket payload section and test tolerances between the payload wall and CanSat.
Vaccum Test	• Construct a vacuum chamber and simulate air pressure changes to ensure all systems and flight software function correctly.

13.4.5 Simulation Test Plan

GCS Simulation Test Plan		
Test Category	Procedure of Test	Requirement to Pass
Ground Control Station, Sensors, Communications	Power on CanSat and GS computer. Send Activate/Enable simulation commands, then transmit simulated pressure data at 1Hz using the SIMP command.	Received packets must contain mission guide telemetry fields with correct resolution. GS must send/receive telemetry at 1Hz with proper graphics and CSV logging.

FSW Simulation Test Plan		
Test Category	Procedure of Test	Requirement to Pass
Flight Software, Sensors, Camera, Communications	Verify GCS barometric pressure data generates altitude. Enable simulation mode (ENABLE command), then start the simulation (ACTIVATE command) to stop real sensor pressure reading.	CanSat enters Simulation Mode as commanded. It must receive simulated pressure data, calculate altitude, and activate onboard components like the camera per flight states.



14 Management & Logistics

14.1 Electronics and Sensor Procurement

S No.	Component	Quantity	Status	Date Ordered	Date Received	Purpose
1	SIM808 GSM/GPRS/GPS Module	1	Procured	11 Feb 2025	14 Feb 2025	N/A
2	Pressure Sensor LPS22HB	1	Procured	11 Feb 2025	14 Feb 2025	Pressure
3	MPU9250	1	Procured	11 Feb 2025	15 Feb 2025	Tilt and Rotation
4	BME680	1	Procured	11 Feb 2025	15 Feb 2025	Redundancy
7	ADS1115	1	In Stock	N/A	N/A	Voltage
8	AHT21B	1	In Stock	N/A	N/A	Temperature
9	DS3231	1	In Stock	N/A	N/A	Real time clock
10	BMP390	1	In Stock	N/A	N/A	Redundancy
11	PX4	1	In Stock	N/A	N/A	N/A
12	MS 4525DO	1	N/A	N/A	N/A	Air Speed
13	NEO-M8N	1	N/A	N/A	N/A	GPS Position
14	M135	1	N/A	N/A	N/A	Redundancy
15	Prana Pas 01	1	In Stock	N/A	N/A	Outdoor Air Quality
16	MQ131	1	N/A	N/A	N/A	Redundancy

14.2 Mechanical Procurement

S No.	Component	Quantity	Status	Date Ordered	Date Received
1	Carbon Fibre Sheet Plate	3	Not Yet Procured	N/A	N/A
2	Carbon Fibre Hollow Tubes	2	Procured	11 Feb 2025	15 Feb 2025
3	Acrylic Sheets	2	Procured	11 Feb 2025	14 Feb 2025
4	Screws and nuts	-	Procured	11 Feb 2025	14 Feb 2025
5	Ripstop Nylon	-	Procured	14 Feb 2025	15 Feb 2025
6	3-D Printed Parts	-	Not Yet Procured	N/A	N/A

14.3 Bill of Materials – Weight Summary

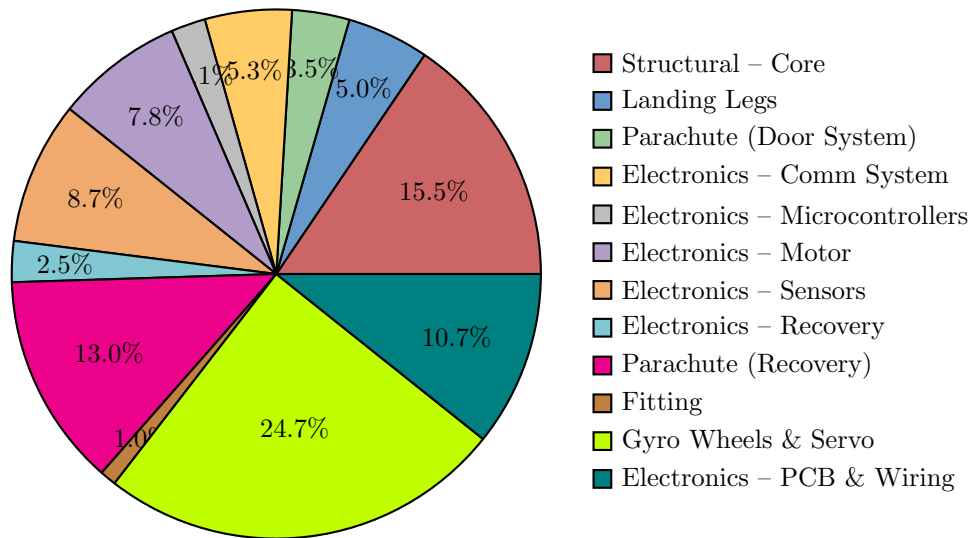
Category	Component	Details	Quantity	Weight Each (g)	Total Weight (g)
Structural – Core					
	Plates	Carbon fiber	5	10	50
	Base plate	ABS	1	10	10
	Main rods	Carbon fiber	3	5	15
	End fitting	Steel	2	5	10
	Plate spacers	Carbon fiber tubes	3	15	45
	Side panels	PLA	3	10	30
Subtotal – Structural					160 g
Landing Legs					
	Landing Leg Component	(2 rods + 1 fitting per leg; 4 legs total)	12	4.3	52 g
Subtotal – Landing Legs					52 g
Parachute (Door System)					
	Door	PLA	1	10	10
	Spring system	Steel springs	3	5	15
	Door latch	Plastic	1	11	11
Subtotal – Parachute (Door)					36 g
Electronics – Comm System					
	Xbee S2C pro	Transmitter module on CanSat	1	5	5
	GSM	—	1	35	35
	Antenna for XBee	—	1	15	15
<i>Continued on next page</i>					

Category	Component	Details	Quantity	Weight Each (g)	Total Weight (g)
Subtotal – Comm System					55 g
Electronics – Microcontrollers					
	ESP32	—	1	12	12
	MicroSD shield	—	1	5	5
	RTC	—	1	5	5
Subtotal – Microcontrollers					22 g
Electronics – Motor					
	N200 Motor	(Gyro motor)	3	20	60
	TB6612FNG	Motor Driver	2	10	20
Subtotal – Electronics – Motor					80 g
Electronics – Sensors					
	Ublox Neo M8N	(GPS)	1	30	30
	BME680	(pressure, humidity, temp, VOC)	1	7	7
	MPU9250	(IMU)	1	8	8
	Prana Air Outdoor	(PM2.5)	1	20	20
	MQ135	(NO _x , CO ₂)	1	10	10
	MQ131	(Ozone)	1	10	10
	ADS1115	(Voltage)	1	5	5
	High G accelerometer	On hold	1	TBD	TBD
Subtotal – Sensors					90 g
Electronics – Recovery					
	RS Pro Buzzer	92 dB alert buzzer	1	20	20
Continued on next page					

Category	Component	Details	Quantity	Weight Each (g)	Total Weight (g)
	LED Indicators	Assorted colors	3	2	6
Subtotal – Electronics Recovery					26 g
Parachute (Recovery)					
	Main chute	Ripstop nylon, Dia: 1.5m	1	120	120
	Drogue chute	Ripstop nylon, Dia: 25cm	1	4.1	4.1
	Cording	Paracord with swivel	1	10	10
Subtotal – Parachute (Recovery)					134.1 g
Fitting					
	Nuts + screws	—	1	10	10
Subtotal – Fitting					10 g
Gyro Wheels & Servo					
	Gyro Wheel	(Wheel only)	3	75	225
	Servo	—	1	30	30
Subtotal – Gyro Wheels & Servo					255 g
Electronics – PCB & Wiring					
	PCB Assembly & Wiring	—	1	110	110
Subtotal – PCB & Wiring					110 g
Overall Known Total:					1030.1 g

*Totals exclude items marked as TBD/—.

14.3.1 Weight Distribution by Subsystem



14.4 Bill of Materials – Cost Summary (in Rs)

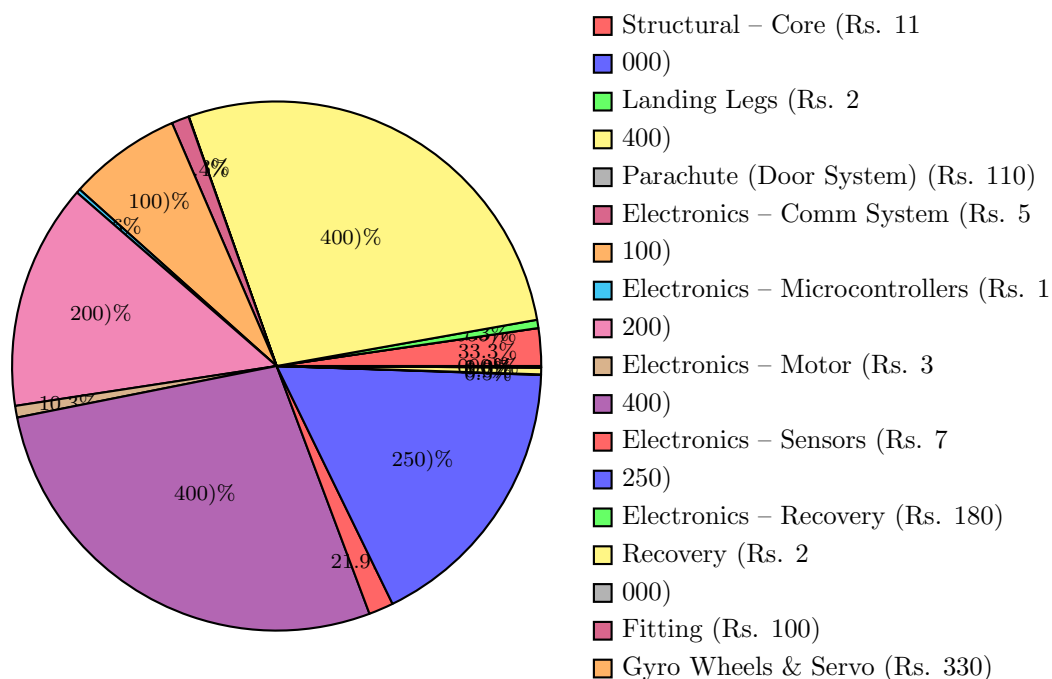
Category	Component	Details	Quantity	Cost Each (Rs)	Total Cost (Rs)
Structural – Core					
	Plates	Carbon fiber	5	200	1,000
	Base plate	ABS	1	200	200
	Main rods	Carbon fiber	3	2,000	6,000
	End fittings	Steel	6	100	600
	Plate spacers	Carbon fiber, 1cm height	100	20	2,000
	Side panels	PLA	3	400	1,200
Subtotal – Structural					11,000 Rs
Landing Legs					
	Landing Leg Component	Combination of rod and fitting	12	200	2,400
Subtotal – Landing Legs					2,400 Rs
Parachute (Door System)					
	Door	—	1	100	100
	Spring system	—	3	—	—
	Door latch	—	1	10	10
Subtotal – Parachute (Door)					110 Rs*
Electronics – Comm System					
	Xbee 3 pro	(1 on CanSat, 1 on ground)	1	3,500	3,500
	GSM	—	1	1,200	1,200
	Antenna for XBee	—	1	400	400
Subtotal – Comm System					5,100 Rs
Electronics – Microcontrollers					
<i>Continued on next page</i>					

Category	Component	Details	Quantity	Cost Each (Rs)	Total Cost (Rs)
	ESP32	—	1	1,000	1,000
	MicroSD shield	—	1	100	100
	RTC (DS3231)	—	1	100	100
Subtotal – Microcontrollers					1,200 Rs
Electronics – Motor					
	N200	Motor	3	1,000	3,000
	TB6612FNG	Motor Driver	2	200	400
Subtotal – Motors					3,400 Rs
Electronics – Sensors					
	Ublox Neo M8N	(GPS)	1	1,000	1,000
	BME680	(pressure, humidity, temp, VOC)	1	600	600
	MPU9250	(IMU)	1	500	500
	Prana Air Outdoor	(PM2.5)	1	1,500	1,500
	MQ135	(NO _x , CO ₂)	1	300	300
	BMP390	(pressure)	1	1,000	1,000
	MQ131	(Ozone)	1	300	300
	ADS1115	(Voltage)	1	300	300
Subtotal – Sensors					5,500 Rs
Electronics – Recovery					
	RS Pro Buzzer	92 dB alert buzzer	1	150	150
	LED Indicators	Assorted colors	3	10	30
Subtotal – Electronics Recovery					180 Rs
Recovery					
Continued on next page					

Category	Component	Details	Quantity	Cost Each (Rs)	Total Cost (Rs)
	Main chute	Ripstop nylon, Dia: 1.5m	1	1,000	1,000
	Drogue chute	Ripstop nylon, Dia: 25cm	1	500	500
	Cording	Paracord with swivel	1	500	500
Subtotal – Recovery					2,000 Rs
Fitting					
	Nuts + screws	—	1	100	100
Subtotal – Fitting					100 Rs
Overall Known Total:					30,990 Rs

*Totals exclude items marked as TBD/—.

14.4.1 Cost Distribution by Subsystem



Cost Distribution by Subsystem

14.5 Location and Recovery Plans

We plan to investigate multiple ways in which we can locate the CanSat:

14.5.1 Audible Buzzer (>92 dB)

We have selected the **RS PRO 92dB Panel Mount Continuous Internal Buzzer** as it meets our operational requirements. This buzzer operates within a voltage range of **3-20V**, with a sound level of **92 dB**, ensuring sufficient audibility for recovery purposes. By adjusting the voltage, we can generate a pulsing tone that enhances distinctiveness, making it less likely to blend with ambient noise. The cost of the buzzer is **INR 700/-**, and further technical details can be found in its [datasheet](#).

14.5.2 Light

Four pulsing LEDs will be used for early morning recovery. We will choose wavelengths that are not present in the environment naturally to improve visibility.

14.5.3 GSM/GPRS

Since the XBEE signal is probably lost due to the lack of line of sight, GSM/GPRS can still function effectively. This method does not require full line-of-sight communication and has a higher peak transmission power, enabling location data transmission and additional packets. It also serves as a backup to the GPS system, which can fail if obstructed.

14.5.4 Radio Beacon

We plan to integrate a simple radio beacon that emits a coded frequency pulse at regular intervals, allowing us to locate the CanSat. More details on construction techniques can be found [here](#).

14.5.5 WiFi-Based Tracking

The ESP32 onboard WiFi will be configured to broadcast on a specific channel, allowing us to detect and locate the CanSat using a WiFi scanner on a mobile phone. This method is effective in non-cluttered environments.

14.5.6 Bright Fluorescent Colors

The 3D-printed parts will be coated with a reflective clear layer to improve visibility. If constraints prevent coloring the structure itself, we will add a coat of abrasion-resistant clear paint to enhance visibility. Additional information is available [here](#).

14.6 Shipping and Travelling

The CanSat team, based in Hyderabad, benefits from excellent connectivity to most parts of India via both rail and air transport. If Bangalore is designated as the launch site, there are multiple travel options available, including the Vande Bharat Express, which operates almost daily, frequent flights between Hyderabad and Bangalore, and bus services as an alternative. For transportation of the CanSat and essential tools required for assembly and troubleshooting, a portion of the equipment will be shipped to a designated location at the launch site, such as a hotel or a contact's residence, while the remaining items will be carried by the team members. The travel expenses will be borne by the team members individually and will not be included in the overall budget for the development of the CanSat.

14.7 Detailed Program Schedule

Administration	Team Members	Start Date	End Date	No. Of Days	Progress (%)
Team Member Recruitment	Team	06/01/2024	06/30/2024	30	100
First meeting of the team	Team	09/01/2024	09/01/2024	1	100
Analysis of the Mission Guide	Team	09/02/2024	09/06/2024	5	100
Prepare the first draft of Proposal	Team	09/07/2024	09/30/2024	24	100
Intra-College Submission of CanSat Proposal	Team	10/01/2024	10/02/2024	2	100
Merging of Intra College Teams	Team	10/03/2024	10/03/2024	1	100
First combined Meeting	Team	10/04/2024	10/04/2024	1	100
Sub team Division and Task Allocation	Team	10/04/2024	10/14/2024	11	100
Preparing the PDR	Team	10/15/2024	01/15/2025	92	100
End Sem Examinations		11/23/2024	12/01/2024	9	100
Academic Work and Project Submissions		12/01/2024	12/06/2024	6	100
Winter Break		12/06/2024	01/01/2025	27	100

Administration	Team Members	Start Date	End Date	No. Of Days	Progress (%)
Exploring Colleges Resources Available	Team	12/20/2024	12/25/2024	6	70
Internal Funding	Team	12/20/2024	12/25/2024	6	100
Quiz Examinations		01/29/2025	01/31/2025	4	100
Submit the PDR	Team	01/15/2025	01/15/2025	1	100
Preparing the CDR	Team	01/16/2025	03/05/2025	49	100
Mid Semester Break		02/26/2025	02/29/2025	4	100
1st Examinations week		02/15/2025	03/01/2025	14	100
2nd Examinations week		03/25/2025	03/31/2025	7	80
Acquiring University Lab Permissions for Final Testing	Team	03/22/2025	04/09/2025	18	70
Submit the CDR	Team	03/05/2025	03/05/2025	1	100
3D Printing of Final Build	Mech Team	09/01/2025	09/10/2025	10	20
PCB Printing the Final Circuits	Team	09/11/2025	09/14/2025	4	0
End Semester Exams		08/25/2025	09/05/2025	12	0
Assembling the final CanSat	Team	09/15/2025	09/21/2025	7	10
Performing Integrated Tests	Team	09/22/2025	09/26/2025	5	0
Repairing CanSat	Team	09/27/2025	09/29/2025	3	0
Performing 2nd Phase Tests	Team	09/30/2025	10/03/2025	4	0
Repairing CanSat	Team	10/03/2025	10/05/2025	3	0
Performing 3rd Phase Tests	Team	10/06/2025	10/08/2025	3	0
Compiling Results and Preparing Environmental Test Document	Team	10/09/2025	10/13/2025	5	0

Administration	Team Members	Start Date	End Date	No. Of Days	Progress (%)
Mission Rehearsals	Team	10/14/2025	10/21/2025	8	0
Launch Weekend	Team	10/22/2025	10/25/2025	4	0

15 Conclusion

The following section summarizes the progress made so far and outlines the next steps for the team as we move forward in the competition.

15.1 Major Accomplishments

Throughout the development of our CANSAT, we have successfully completed several key milestones across multiple subsystems:

- **Mechanical:** All simulations have been completed, major mechanisms tested, and all major parts have been 3D printed. Prototyping has been conducted and validated.
- **Hardware:** All sensors have been tested and integrated, and XBee communication tests have been largely completed. Work on our Gyroscope is in progress.
- **Software:** GUI design and simulation modes have been implemented, processor resets handled, and communication with sensors successfully integrated.
- **Administration and Sponsorship:** Sponsorship and partnership opportunities have been explored, and initial contacts have been established.

15.2 Major Unfinished Work

While significant progress has been made, certain areas require further development and refinement:

- **Mechanical:** Assembly of the CANSAT and overall system integration.
- **Hardware:** Gyroscope requires further testing and refinements if possible
- **Software:** Rigorously testing the control algorithm and
- **Administration and Sponsorship:** Securing sponsorship confirmations, and travel bookings.

15.3 What's Next?

Team Brahmand has successfully navigated crucial design, prototyping, and manufacturing stages, remaining aligned with our planned timeline and competition deadlines. Moving forward, our focus will shift towards extensive and rigorous testing to validate system performance and ensure mission success.

With continued dedication and teamwork, we are well-prepared to advance to the next stage of development, refining our systems for a seamless launch and deployment.

16 Acknowledgment

The successful completion of this **Complete Design Review (CDR) Document** for the INSPACE CANSAT Competition would not have been possible without the guidance, support, and dedication of numerous individuals and organizations.

We extend our deepest gratitude to **INSPACE** for organizing this competition and providing us with an invaluable opportunity to apply our engineering knowledge in a real-world aerospace challenge. Their continuous efforts in fostering innovation and hands-on learning have been instrumental in shaping our understanding of satellite technology and space systems.

We would also like to express our sincere appreciation to our **mentors, professors, and faculty advisors** for their unwavering support, insightful guidance, and technical expertise. Their encouragement and constructive feedback have played a pivotal role in refining our design and ensuring the robustness of our CANSAT system.

A special thanks to the **teammates** for their dedication, collaboration, and perseverance throughout this project. The synergy of our diverse skills and expertise has allowed us to overcome challenges, develop innovative solutions, and work towards a common goal.

We are also grateful to our **institutions, sponsors, and industry professionals** who have provided us with resources, infrastructure, and funding support. Their contributions have significantly enhanced our ability to conduct research, develop prototypes, and perform rigorous testing.

Lastly, we acknowledge the support and encouragement of our **families and friends**, whose patience and motivation have been our pillars of strength throughout this journey.

Thanking You,

Team Brahmand,

IIIT Hyderabad

17 Appendix

17.1 Datasheet References with Updated Links

Samsung 18650-30Q Li-ion Battery:

Product page: <https://robu.in/product/samsung-18650-30q-li-ion-battery/>

BME680 (Temperature, Humidity, Pressure, and Gas Sensor):

Smartelex BME680 Breakout Board

Product page: <https://robu.in/product/smartelelex-bme680-temperature-humidity-pressure-and-gas-sensor-breakout-board/>

Winsen MQ131 High Concentration Ozone Gas Sensor:

Product page: <https://robu.in/product/winsen-mq131-high-concentration-ozone-gas-sensor/>

Winsen MQ135 Semiconductor Sensor for Air Quality:

Product page: <https://robu.in/product/winsen-mq135-semiconductor-sensor-for-air-quality/>

Prana Air Outdoor PM Sensor (PM 2.5 Sensor):

Product page: <https://www.pranaair.com/us/air-quality-sensor/outdoor-pm-sensor/>

Smartelex LPS22HB Pressure Sensor:

Product page: <https://robu.in/product/smartelelex-lps22-pressure-sensor-lps22hb/>

MPU9250 (9-axis Attitude, Gyro, Accelerometer, Magnetometer Sensor Module):

Product page: <https://robu.in/product/mpu9250-9-axis-attitude-gyro-accelerator-magnetometer-sensor-module/>

LM2596S DC-DC Buck Converter Power Supply:

Product page: <https://robu.in/product/lm2596s-dc-dc-buck-converter-power-supply/>

AHT21B Temperature and Humidity Sensor:

Product page: <https://robu.in/product/temperature-and-humidity-sensor-model-aht21b/>